Classification of a class of quantum graphs over complex 3×3 matrices

(Klasyfikacja pewnej klasy grafów kwantowych
nad zespolonymi macierzami 3×3)

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Master's thesis

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Abstract

According to all known laws of aviation, there is no way a bee should be able to fly. Its wings are too small to get its fat little body off the ground. The bee, of course, flies anyway because bees don't care what humans think is impossible.

Według wszystkich znanych praw lotnictwa, pszczoła nie powinna w ogóle latać. Jej skrzydła są zbyt małe, by unieść jej pulchne ciałko z ziemi. Pszczoła, oczywiście, lata mimo wszystko, ponieważ pszczoły nie przejmują się tym, co ludzie uważają za niemożliwe.

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Chapter 1

Introduction

1.1 Motivation

Quantum graphs is a topic that has seen a resurgence in interest in recent years.

1.2 Scope

This thesis is organised as follows.

We begin, in the preliminary chapter, by recalling the algebraic background required for everything that follows: dual groups and self-duality of finite abelian groups, positive and faithful functionals on C^*-algebras, group actions and crossed products, and the theory of matrix Lie groups and Lie algebras – in particular the unitary groups $U(n)$, $SU(n)$, and $PU(n)$, together with their associated Lie algebras $\mathfrak{gl}_n$, $\mathfrak{sl}_n$, and $\mathfrak{su}_n$, which will serve throughout as the spaces in which quantum graphs are classified.

Building on this foundation, we then introduce the central objects of the thesis: *quantum sets* and *quantum graphs*, following Schäfer and Tobolski [15]. We show how classical sets and matrix algebras both give rise to quantum sets, define quantum graphs as Schur-idempotent, $*$-preserving maps generalising classical adjacency matrices, and extend the classical theory of voltage and derived graphs to this setting. A key result shows that derived quantum graphs built from different group actions on the same voltage data are always quantum isomorphic, a tool that will later let us relate quantum graphs on $M_3(\mathbb{C})$ back to classical graphs. We close the chapter by recalling the notion of *quantum isomorphism*, and by stating Gromada's theorem, which translates quantum graphs on $M_n(\mathbb{C})$ into an explicit correspondence with linear subspaces of $\mathfrak{gl}_n$.

The remainder of the thesis is devoted to quantum graphs over 3×3 matrices. Using the generalised Pauli (clock-and-shift) matrices P, Q, we construct and completely classify, up to unitary equivalence, all loop-free, undirected quantum graphs on $M_n(\mathbb{C})$ built from this family, showing that they fall into exactly five isomorphism classes. We then realise each nontrivial class concretely as the derived quantum graph of an explicit classical voltage graph over $\widehat{\mathbb{Z}}_3$, and show that every such graph is vertex-transitive. Finally, we show that the clean, single-invariant classification available for $M_2(\mathbb{C})$ – resting on the exceptional isomorphism $PU(2) \cong SO(3)$ – provably fails to generalise to $M_3(\mathbb{C})$, where infinite families of mutually non-isomorphic quantum graphs exist for each fixed number of quantum edges, and where quantum graphs can be exhibited that are not quantum isomorphic to any classical graph at all.

1.3 Acknowledgements

I would like to thank my supervisor Mariusz Tobolski for his unwavering support and patience.

Chapter 2

Preliminaries

The following sections cover various topics in algebra. They are not intended to provide an exhaustive treatment; rather, they are meant to equip the reader with the necessary background to understand the remainder of this work. References to further information are provided where appropriate.

2.1 Group theory

In the following section, we will introduce some group theoretic notions that be used throughout this thesis. We assume that the reader is familiar with the notion of a group. We denote the torus by $\mathbb{T}$, i.e. the set of complex numbers with the absolute value of 1, along with multiplication as its group action. We denote the identity element of G by e_G. If G is abelian, its identity element is expressed by 0_G.

The following definition and proposition will be used throughout the thesis, as almost all of the groups we will be dealing with will be finite and abelian.

Definition 2.1.1 (Dual of a group, Rudin [14], Section 1.2)**.** Let G be a finite abelian group. The *dual group* $\widehat{G}$ is the set of all homomorphisms from G into the torus $\mathbb{T}$ with the group operation defined as

$$(\varphi\psi)(g) = \varphi(g)\psi(g)$$

for all $\varphi, \psi \in \widehat{G}$ and $g \in G$, where the latter multiplication takes place in $\mathbb{C}$.

Remark 2.1.1. We will often call the elements of the dual group *characters.* If a group is isomorphic to its dual it will be called *self-dual.*

Proposition 2.1.1. *If G is finite and abelian, then it is self-dual.*

Proof. First we will prove the statement for finite cyclic groups. Let $G = \langle g \rangle$, $\mathrm{ord}(g) = n$. We will prove that the dual is cyclic of order n. Let $\chi\colon G \to \mathbb{T}$

be a character. Since the group is cyclic, the character is entirely determined by the value it takes on the generator g. Since $g^n = e$, obviously $\chi(g)^n = 1$, so $\chi(g) \in \mu_n = \{\exp(2k\pi i/n) : k = 0, 1, \ldots, n-1\} \subseteq \mathbb{T}$. For each possible choice of $\chi(g) \in \mu_n$ we get a different character. It is then easy to see that $\chi : G \to \mathbb{T}$ such that $\chi(g) = \exp(2\pi i/n)$ generates this group, i.e. $\widehat{G} = \{\chi, \chi^2, \ldots, \chi^{n-1}, \chi^n = e_{\widehat{G}}\}$.

By the fundamental theorem of finite abelian groups (see Dummit and Foote [5, Section 5.2]), since every finite abelian group can be written as a finite product of finite cycles, all we need to do is check whether the statement holds for a product of two groups A, B that are themselves self-dual, i.e. $\widehat{A \times B} \cong \widehat{A} \times \widehat{B}$. This is self-evident as every character $\chi \in \widehat{A \times B}$ is determined by its restrictions to $A \times \{e_B\}$ and $\{e_A\} \times B$, since

$$\chi(a, b) = \chi((a, e_B)(e_A, b)) = \chi(a, e_B)\chi(e_A, b).$$

Thus $\chi \leftrightarrow (\chi|_A, \chi|_B)$ is an isomorphism. □

Remark 2.1.2. While every finite abelian group G is self-dual, there is no canonical isomorphism $G \cong \widehat{G}$: the isomorphism constructed in the proof of Proposition 2.1.1 depends on a choice of generator for each cyclic factor, and different choices yield different isomorphisms. This is in sharp contrast with the *double* dual $\hat{\hat{G}}$, which is canonically isomorphic to G via the evaluation map $g \mapsto (\chi \mapsto \chi(g))$, with no arbitrary choices involved.

This distinction is the reason we consistently work with the dual group $\widehat{G}$, rather than G itself, when defining actions on quantum sets and crossed products later in this thesis (see e.g. Definition 3.1.2). A group action $\alpha\colon G \to \mathrm{Aut}(B, \psi)$ and its *dual* counterpart $\widehat{\alpha}\colon \widehat{G} \to \mathrm{Aut}(B, \psi)$ are related by an isomorphism $G \cong \widehat{G}$, but since this isomorphism is not canonical, working with $\widehat{\alpha}$ directly avoids committing to an arbitrary identification.

2.2 C^*-algebras

In the following section, we will introduce some notions stemming from the theory of C^*-algebras. We assume the reader is familiar with the notion of a C^*-algebra. The following definitions are taken from Blackadar [2].

Definition 2.2.1 (Banach and C^*-algebras, Definition II.1.1.1)**.** A *Banach algebra* is a complex algebra which is a Banach space under a norm that is submultiplicative, meaning

$$\|ab\| \leq \|a\|\|b\| \quad \text{for all } a, b \in A.$$

An *involution* on a Banach algebra A is a conjugate-linear isometric antiautomorphism of order two, usually denoted $a \mapsto a^*$. In other words, for all $a, b \in A$ and $\lambda \in \mathbb{C}$, it satisfies:

$$(a + b)^* = a^* + b^*, \quad (ab)^* = b^*a^*, \quad (\lambda a)^* = \bar{\lambda}a^*, \quad (a^*)^* = a, \quad \|a^*\| = \|a\|.$$

A *Banach $*$-algebra* is a Banach algebra equipped with an involution. A Banach algebra A is called *unital* if there is an element 1_A such that $1_A a = a 1_A = a$ for all $a \in A$. Such an element will be called an *unit*.

An (abstract) *C^*-algebra* is a Banach $*$-algebra A satisfying the C^*-axiom:

$$\|a^*a\| = \|a\|^2 \quad \text{for all } a \in A.$$

Remark 2.2.1. Throughout this thesis we will assume that C^*-algebras are unital.

Definition 2.2.2 (C^*-algebra representation, II.6.1.1)**.** Let A be a C^*-algebra. A *C^*-algebra representation* π is a *-homomorphism $\pi : A \to B(\mathcal{H})$ for some Hilbert space $\mathcal{H}$. A C^*-algebra representation is called *unital* if $\pi(1_A) = \mathrm{Id}_{\mathcal{H}}$. It is called *faithful* if it is injective.

Definition 2.2.3 (Tensor Product of Finite-Dimensional C^*-Algebras, Section II.9.1)**.** Let A and B be finite-dimensional C^*-algebras. The *tensor product* $A \otimes B$ is the algebraic tensor product $A \otimes_{\mathbb{C}} B$ equipped with the structure of a C^*-algebra, where the multiplication and involution are defined on simple tensors by

$$(a_1 \otimes b_1)(a_2 \otimes b_2) = a_1a_2 \otimes b_1b_2, \qquad (a \otimes b)^* = a^* \otimes b^*,$$

and extended to all of $A \otimes B$ by linearity, and the norm is defined for any $x \in A \otimes B$ by

$$\|x\| = \|(\pi_A \otimes \pi_B)(x)\|_{B(\mathcal{H}_1 \otimes \mathcal{H}_2)},$$

where $\pi_A \colon A \to B(\mathcal{H}_1)$ and $\pi_B \colon B \to B(\mathcal{H}_2)$ are faithful $*$-representations of A and B on finite-dimensional Hilbert spaces $\mathcal{H}_1$ and $\mathcal{H}_2$.

Remark 2.2.2. Because $A \otimes B$ is finite-dimensional, it is automatically complete under this norm and admits a unique C^*-norm independent of the choice of faithful representations.

Remark 2.2.3. Multiplication in a C^*-algebra A is a bilinear map

$$m \colon A \times A \to A.$$

By the universal property of the tensor product, m extends uniquely to a linear map

$$\widetilde{m} \colon A \otimes A \to A.$$

We will not distinguish between m and $\widetilde{m}$ and simply write m, relying on context to distinguish between the two.

Definition 2.2.4 (Linear functional, II.6.2.1)**.** A *linear functional* on a C^*-algebra A is a linear function $\psi : A \to \mathbb{C}$.

Remark 2.2.4. We will call linear functionals simply as *functionals*.

The following notions will be useful for defining an inner product on C^*-algebras. Positiveness and faithfulness are all we need to define a norm on a finite-dimensional C^*-algebra equipped with a suitable functional.

Definition 2.2.5 (Positive functional, II.6.7.1)**.** A functional ψ on a C^*-algebra A is called *positive*, if

$$0 \leq \psi(a^*a)$$

for any $a \in A$.

Definition 2.2.6 (Faithful functional, II.6.7.1)**.** A positive functional ψ on a C^*-algebra A is called *faithful*, if

$$\psi(a^*a) = 0 \Leftrightarrow a = 0$$

for any $a \in A$.

Remark 2.2.5. In the literature, a normalised functional ψ such that $\psi(1) = 1$ is often called a *state*.

Example 2.2.1 (Positive and faithful functionals on matrix algebras)*.* Let $A = M_n(\mathbb{C})$, the C^*-algebra of $n \times n$ complex matrices, equipped with the usual trace Tr. Define

$$\psi(a) = n\,\mathrm{Tr}(a), \qquad a \in A.$$

Since a^*a is positive semi-definite for every $a \in A$, its eigenvalues are non-negative, and hence $\mathrm{Tr}(a^*a) \geq 0$; thus ψ is a positive functional. Moreover, $\mathrm{Tr}(a^*a) = 0$ forces all eigenvalues of a^*a to vanish, which (since a^*a is positive semi-definite, hence diagonalizable with non-negative eigenvalues) implies $a^*a = 0$ and therefore $a = 0$. So ψ is faithful.

Example 2.2.2 (A positive functional that is not faithful)*.* Let $A = \mathbb{C}^2 = \mathbb{C} \oplus \mathbb{C}$, with coordinatewise multiplication and involution given by complex conjugation in each coordinate. Define

$$\psi(x, y) = x, \qquad (x, y) \in A.$$

For $(x, y) \in A$ we have $(x, y)^*(x, y) = (|x|^2, |y|^2)$, so $\psi\big((x, y)^*(x, y)\big) = |x|^2 \geq 0$, and ψ is positive. However $\psi\big((0, y)^*(0, y)\big) = 0$ for every $y \in \mathbb{C}$, while $(0, y) \neq 0$ whenever $y \neq 0$. Hence ψ is positive but *not* faithful.

The following notion is akin to the notion of a group action on a set.

Definition 2.2.7 (Discrete group action on a C^*-algebra)**.** By *group action* of a discrete group G on a C^*-algebra A we mean a group homomorphism

$$\alpha\colon G \to \mathrm{Aut}(A).$$

We will write $g \cdot a$ instead of $\alpha(g)(a)$ then the action in question is clear.

Example 2.2.3 (Group actions on C^*-algebras)*.*

1. *Trivial action.* For *any* C^*-algebra A and *any* group G, setting $\mathrm{triv}(g)(a) = a$ for all $g \in G$, $a \in A$ trivially defines a group action, since $\mathrm{triv}(g) = \mathrm{Id}_A \in \mathrm{Aut}(A)$ for every g, and the identity map is trivially a group homomorphism from G into $\mathrm{Aut}(A)$.

2. *Permutation action.* Let $A = M_n(\mathbb{C})$ and let $G = S_n$, the symmetric group on n letters. For $\sigma \in S_n$, let P_σ be the associated permutation matrix, i.e. $P_\sigma e_i = e_{\sigma(i)}$ on the standard basis $\{e_i\}$ of $\mathbb{C}^n$. Then G acts on A via

$$\alpha(\sigma)(a) = P_\sigma a P_\sigma^{-1}, \qquad \sigma \in S_n,\ a \in A,$$

i.e. $\alpha(\sigma)$ simultaneously permutes the rows and columns of a according to σ. As P_σ is unitary (with $P_\sigma^{-1} = P_\sigma^*$), each $\alpha(\sigma)$ is a $*$-automorphism of A, and since $\sigma \mapsto P_\sigma$ is itself a group homomorphism ($P_{\sigma\tau} = P_\sigma P_\tau$), we get $\alpha(\sigma\tau) = \alpha(\sigma)\alpha(\tau)$, so $\alpha\colon S_n \to \mathrm{Aut}(A)$ is indeed a group homomorphism.

3. *Translation action.* Let $A = C(\mathbb{T})$, the continuous complex-valued functions on the torus $\mathbb{T}$, and let $G = \mathbb{Z}$ act by rotation:

$$(n \cdot f)(z) = f(e^{-in\theta} z), \qquad n \in \mathbb{Z},\ f \in A,$$

for a fixed $\theta \in \mathbb{R}$. This is precisely the action used to construct the irrational rotation algebra as a crossed product (see Example 2.2.5).

2.2.1 Crossed product C^*-algebra

In this subsection, we introduce the notion of a crossed product C^*-algebra, in the limited form needed for our purposes; for a more general treatment we refer the reader to Blackadar [2]. The definitions below follow Blackadar [2].

Definition 2.2.8 (C^*-dynamical system, II.10.3.1)**.** A *C^*-dynamical system* is a triple (A, G, α), where A is a unital C^*-algebra, G is a discrete group, and $\alpha\colon G \to \mathrm{Aut}(A)$ is a group action on A. For a C^*-dynamical system (A, G, α) we set

$$C_c(G, A) = \{f\colon G \to A : f \text{ has finite support}\},$$

the space of all A-valued functions on G with finite support. We usually write $g \cdot a$ for $\alpha(g)(a)$.

If G is finite, every $f\colon G \to A$ is automatically finitely supported. For each $g \in G$, let $u_g \in C_c(G, A)$ denote the function

$$u_g(h) = \begin{cases} 1_A, & h = g, \\ 0_A, & h \neq g, \end{cases} \qquad h \in G,$$

i.e. the indicator function of $\{g\}$ valued in the unit of A. Given $f \in C_c(G, A)$, write $a_g := f(g) \in A$ for its values; since $a_g\, u_g(h) = \delta_{g,h}\, a_g$ (using the multiplication of A and that $u_g(h)$ is either 1_A or 0_A), f is recovered uniquely as

$$f = \sum_{g \in G} a_g\, u_g,$$

where the sum is finite since only finitely many a_g are nonzero. In this way $\{u_g\}_{g\in G}$ plays the role of a formal basis of $C_c(G, A)$ as an A-module, with each u_g concretely realized as the A-valued indicator function of g above.

The space $C_c(G, A)$ carries both a product and an involution, defined precisely so that the group action becomes *inner*, i.e. so that $g \cdot a = u_g a u_g^*$ for all $g \in G$, $a \in A$. Explicitly, for $a = \sum_{g\in G} a_g u_g$ and $b = \sum_{h\in G} b_h u_h$ in $C_c(G, A)$, the product and involution are given by

$$ab = \sum_{g,h\in G} a_g\, (g \cdot b_h)\, u_{gh}, \qquad a^* = \sum_{g\in G} (g^{-1} \cdot a_g^*)\, u_{g^{-1}}.$$

Definition 2.2.9 (Covariant representation, II.10.3.4)**.** Fix a dynamical system (A, G, α). A *covariant representation* $(\pi, u, \mathcal{H})$ of (A, G, α) consists of a unitary representation $u\colon G \to B(\mathcal{H})$ of G and a representation $\pi\colon A \to B(\mathcal{H})$ of A such that

$$u(g)\, \pi(a)\, u(g)^* = \pi(g \cdot a)$$

for every $g \in G$ and $a \in A$.

Definition 2.2.10 (Full C^*-algebra norm on $C_c(G, A)$)**.** The *full C^*-algebra norm* on $C_c(G, A)$ is given by

$$\|\cdot\| = \sup \|(\pi \times u)(\cdot)\|,$$

where the supremum ranges over all covariant representations $(\pi, u, \mathcal{H})$ of (A, G, α).

Definition 2.2.11 (Crossed product C^*-algebra)**.** The *crossed product* of a C^*-algebra A and a discrete group G is the completion of $C_c(G, A)$ with respect to the full norm. We denote it by

$$A \rtimes_\alpha G.$$

The reader will find some examples of such C^*-algebras below (Examples 2.2.4 and 2.2.5). They are both taken from Blackadar [2].

Example 2.2.4 (Trivial action recovers the group C^*-algebra, II.10.3.15). Let $A = \mathbb{C}$ with the trivial action $\alpha_g = \mathrm{id}$ for all $g \in G$. Then every element of $C_c(G, A, \alpha)$ is just a finitely supported function $G \to \mathbb{C}$, with convolution product and involution reducing to the usual group algebra structure, since the action contributes no twisting. Completing in the full norm yields

$$\mathbb{C} \rtimes_\alpha G \;\cong\; C^*(G),$$

the (full) group C^*-algebra of G. For instance, taking $G = \mathbb{Z}$ gives $\mathbb{C} \rtimes \mathbb{Z} \cong C(\mathbb{T})$, since $C^*(\mathbb{Z})$ is generated by a single unitary with no relations, i.e. the continuous functions on the circle via the Fourier transform.

Example 2.2.5 (Irrational rotation algebra, II.10.4.12). Let $A = C(\mathbb{T})$, the continuous functions on the torus, and let $G = \mathbb{Z}$ act by rotation through angle $2\pi\theta$ for some fixed $\theta \in \mathbb{R} \setminus \mathbb{Q}$, i.e.

$$\alpha_n(f)(z) = f\big(e^{-2\pi i n\theta} z\big), \qquad z \in \mathbb{T},\ n \in \mathbb{Z}.$$

The resulting crossed product

$$A_\theta \ := \ C(\mathbb{T}) \rtimes_\alpha \mathbb{Z}$$

is the *irrational rotation algebra* (a non-commutative torus). It is generated by a unitary u (implementing the $\mathbb{Z}$-action) and a unitary v (generating $C(\mathbb{T})$) subject to the single relation

$$uv = e^{-2\pi i\theta}\, vu,$$

and is a standard first example of a simple, non-commutative C^*-algebra arising from a crossed product.

2.3 Matrix Lie Groups and Algebras

In this section we recall some notions from the theory of matrix Lie groups and algebras that will be needed later in the thesis. The following definitions are taken from Hall [9].

Definition 2.3.1 (Lie group, Definition 1.18)**.** A *Lie group* is a group that is simultaneously a smooth manifold, in such a way that both the group multiplication and the inversion map are smooth.

Definition 2.3.2 (Lie algebra, Definition 2.36)**.** A *Lie algebra* $\mathfrak{g}$ is a vector space over a field (for us, always $\mathbb{C}$ or $\mathbb{R}$) equipped with a bilinear form, the *Lie bracket*,

$$[\cdot,\cdot]\colon \mathfrak{g}\times\mathfrak{g}\to\mathfrak{g},$$

satisfying:

- *antisymmetry*: $[x,y] = -[y,x]$;
- the *Jacobi identity*: $[x,[y,z]] + [y,[z,x]] + [z,[x,y]] = 0$.

Remark 2.3.1 (Corollary 2.35)*.* These two notions are intimately related: every Lie group gives rise to a Lie algebra, obtained as the tangent space at the identity element, together with a bracket induced by the group structure.

Our primary interest lies in *matrix Lie groups*, i.e. subgroups of the general linear group

$$GL(n,\mathbb{C}) = \{A \in M_n(\mathbb{C}) : \det(A) \neq 0\},$$

taken with matrix multiplication as the group operation.

Definition 2.3.3 (Unitary and special unitary groups, Section 1.2.4)**.** The *unitary group* is

$$U(n) = \{U \in M_n(\mathbb{C}) : U^*U = UU^* = I\},$$

and its subgroup, the *special unitary group*, is

$$SU(n) = \{U \in M_n(\mathbb{C}) : U^*U = UU^* = I,\ \det(U) = 1\}.$$

Both carry the subspace topology inherited from $M_n(\mathbb{C})$.

Definition 2.3.4 (Projective unitary group)**.** The *projective unitary group* is

$$PU(n) = U(n)/Z(U(n)),$$

where $Z(U(n))$ denotes the center of $U(n)$. Unlike $U(n)$ and $SU(n)$, the group $PU(n)$ is defined as a *quotient* of $U(n)$ by a normal subgroup, and is not presented here as a subgroup of $GL(n, \mathbb{C})$ in its own right; it carries the quotient topology induced by the projection $U(n) \to U(n)/Z(U(n))$, rather than a subspace topology.

Proposition 2.3.1. *$Z(U(n)) \cong \mathbb{T}$.*

Proof. Recall that the center of $U(n)$ is

$$Z(U(n)) = \{A \in U(n) : AU = UA \text{ for all } U \in U(n)\}.$$

Let $A = (a_{ij}) \in Z(U(n))$. Since A commutes with every unitary matrix, it in particular commutes with every diagonal unitary matrix. Let D_k be the diagonal matrix whose k-th diagonal entry is -1 and all other diagonal entries are 1. Clearly, $D_k \in U(n)$. The condition $AD_k = D_kA$ implies that for all $j \neq k$, the (k, j)-th entry satisfies $-a_{kj} = a_{kj}$, which forces $a_{kj} = 0$. Since this holds for all k, all off-diagonal entries of A must vanish, meaning A is a diagonal matrix.

Furthermore, A also commutes with every permutation matrix, which are orthogonal and hence unitary. Let P_{ij} denote the permutation matrix exchanging the i-th and j-th basis vectors. From

$$AP_{ij} = P_{ij}A$$

we see that conjugating A by P_{ij} leaves it unchanged. Because A is diagonal, conjugation by P_{ij} simply swaps the i-th and j-th diagonal entries. Since this holds for every pair i, j, all diagonal entries of A must coincide. Hence $A = \lambda I_n$ for some $\lambda \in \mathbb{C}$, where by I_n we denote the $n \times n$-identity matrix.

As $A \in U(n)$, we have $A^*A = I_n$, and substituting $A = \lambda I_n$ gives

$$(\overline{\lambda} I_n)(\lambda I_n) = |\lambda|^2 I_n = I_n,$$

so $|\lambda| = 1$.

Conversely, if $A = \lambda I_n$ with $|\lambda| = 1$, then clearly $A \in U(n)$, and for every $U \in U(n)$,

$$AU = (\lambda I_n)U = U(\lambda I_n) = UA,$$

so $A \in Z(U(n))$. This shows that $Z(U(n)) = \{\lambda I_n : \lambda \in \mathbb{C}, |\lambda| = 1\}$.

Finally, the map $\lambda \mapsto \lambda I_n$ is a natural group isomorphism from $\mathbb{T}$ to $Z(U(n))$, which proves the claim. $\square$

Definition 2.3.5 (Special orthogonal group, Section 1.2.3)**.** The *special orthogonal group* is

$$SO(n) = \{R \in M_n(\mathbb{R}) : R^T R = RR^T = I,\ \det(R) = 1\},$$

the Lie group of orientation-preserving rotations fixing the origin.

Proposition 2.3.2 (Matrix commutator)**.** *For matrix Lie algebras, the Lie bracket is given by the matrix commutator*

$$[X, Y] = XY - YX$$

for X, Y in the Lie algebra; this bracket is antisymmetric and satisfies the Jacobi identity.

Proof. Antisymmetry is immediate: $[X, Y] = XY - YX = -(YX - XY) = -[Y, X]$. For the Jacobi identity, expand each term:

$$\begin{aligned}
[X, [Y, Z]] &= X(YZ - ZY) - (YZ - ZY)X = XYZ - XZY - YZX + ZYX,\\
[Y, [Z, X]] &= YZX - YXZ - ZXY + XZY,\\
[Z, [X, Y]] &= ZXY - ZYX - XYZ + YXZ.
\end{aligned}$$

Summing the three lines, every term cancels against its negative counterpart, giving 0. □

Definition 2.3.6 ($\mathfrak{gl}_n$, $\mathfrak{su}_n$, and $\mathfrak{so}_n$, Sections 2.5.2, 2.5.4, and 2.5.5)**.** The Lie algebra associated with $GL(n, \mathbb{C})$ is denoted $\mathfrak{gl}(n, \mathbb{C})$ and consists of *all* complex $n \times n$ matrices,

$$\mathfrak{gl}(n, \mathbb{C}) = M_n(\mathbb{C}).$$

The Lie algebra associated with $SU(n)$, denoted $\mathfrak{su}(n)$, consists of the traceless skew-Hermitian matrices,

$$\mathfrak{su}(n) = \{A \in M_n(\mathbb{C}) : A^* = -A,\ \mathrm{Tr}(A) = 0\}.$$

The Lie algebra associated with $SO(n)$, denoted $\mathfrak{so}(n)$, consists of the skew-symmetric matrices,

$$\mathfrak{so}(n) = \{A \in M_n(\mathbb{C}) : A^T = -A\},$$

where A^T is the matrix transpose.

Proposition 2.3.3. *$\mathfrak{su}(2)$ is a 3-dimensional real vector space.*

Proof. The traceless condition removes one real degree of freedom from the diagonal of a skew-Hermitian matrix, since the diagonal entries of a skew-Hermitian matrix are purely imaginary and their sum must vanish; a 2×2 skew-Hermitian matrix is thus determined by one real diagonal parameter α (with entries $i\alpha, -i\alpha$) together with one free complex off-diagonal entry b, the other being forced to $-\bar{b}$. This gives

$$\mathfrak{su}(2) = \left\{ \begin{pmatrix} i\alpha & b \\ -\bar{b} & -i\alpha \end{pmatrix} : \alpha \in \mathbb{R},\ b \in \mathbb{C} \right\} \cong \mathbb{R} \times \mathbb{C} \cong \mathbb{R}^3$$

as real vector spaces, so $\dim_{\mathbb{R}} \mathfrak{su}(2) = 3$. □

Example 2.3.1 (The circle group $SO(2)$ and its Lie algebra). The group $SO(2)$ consists of rotation matrices

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad \theta \in [0, 2\pi),$$

and is isomorphic, as a Lie group, to the torus $\mathbb{T}$ via $e^{i\theta} \mapsto R_\theta$. The corresponding Lie algebra $\mathfrak{so}(2)$ consists of antisymmetric real matrices,

$$\mathfrak{so}(2) = \left\{ \begin{pmatrix} 0 & -t \\ t & 0 \end{pmatrix} : t \in \mathbb{R} \right\},$$

and is a one-dimensional real vector space. Since any two elements of $\mathfrak{so}(2)$ are scalar multiples of one another, the Lie bracket on $\mathfrak{so}(2)$ vanishes identically: it is an *abelian* Lie algebra.

Example 2.3.2 (A basis for $\mathfrak{su}(2)$). A convenient real basis for $\mathfrak{su}(2)$ is given by $\{i\sigma_1, i\sigma_2, i\sigma_3\}$, where $\sigma_1, \sigma_2, \sigma_3$ are the Pauli matrices

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Each $i\sigma_k$ is traceless and skew-Hermitian, so $i\sigma_k \in \mathfrak{su}(2)$, and the three matrices are linearly independent over $\mathbb{R}$; together with $\dim_{\mathbb{R}} SU(2) = 3$ this confirms that $\mathfrak{su}(2)$ is a 3-dimensional real vector space, as anticipated above. The commutation relations

$$[i\sigma_1, i\sigma_2] = -2i\sigma_3, \qquad [i\sigma_2, i\sigma_3] = -2i\sigma_1, \qquad [i\sigma_3, i\sigma_1] = -2i\sigma_2$$

exhibit $\mathfrak{su}(2)$ as (a rescaling of) the cross-product Lie algebra on $\mathbb{R}^3$, and identify $\mathfrak{su}(2)$ with $\mathfrak{so}(3)$ up to isomorphism.

Chapter 3

Quantum sets and quantum graphs

In this chapter we introduce the notions of quantum sets and quantum graphs. These were first considered independently by Duan, Severini, and Winter [4], in the context of *quantum channels*, and by Weaver [16], in the context of *quantum relations*. Musto, Reutter, and Verdon [12] later developed a rigorous, categorical framework for dealing with quantum sets.

Here we mostly follow the theory developed in Schäfer and Tobolski [15] and Gromada [6].

3.1 Quantum sets

Let B be a C^*-algebra equipped with a faithful positive functional ψ. By the Gelfand–Naimark theorem (Blackadar [2], II.6.4.10), B can be realized concretely as a closed $*$-subalgebra of the bounded linear operators $B(\mathcal{H})$ on some Hilbert space $\mathcal{H}$. When B is finite-dimensional, this abstract picture simplifies considerably: rather than passing to $\mathcal{H}$, we can identify B with its own Hilbert space completion $L^2(B, \psi)$, by equipping B itself with the inner product induced by ψ,

$$\langle x, y \rangle = \psi(x^* y), \qquad x, y \in B.$$

Since ψ is faithful, this form is positive definite, and since B is already finite-dimensional, it is automatically complete; hence $L^2(B, \psi)$ may simply be taken to be B itself, viewed as a Hilbert space via this inner product.

This Hilbert space structure is the key extra piece of data that allows one to speak of adjoints of maps built from B, and in particular of the adjoint of multiplication. This leads to the central definition of this thesis, taken directly from Schäfer and Tobolski [15].

Definition 3.1.1 (Quantum set, Schäfer and Tobolski [15], Definition 2.1)**.** A *quantum set* (B, ψ) is a pair consisting of a finite-dimensional C^*-algebra B and a faithful positive functional $\psi\colon B \to \mathbb{C}$ such that

$$mm^* = \mathrm{Id}_B$$

holds, where $m\colon B \otimes B \to B$ is the multiplication map and m^* is its adjoint with respect to the inner products induced by ψ, i.e. the unique linear map determined by

$$\langle m(a \otimes b), c\rangle = \langle a \otimes b, m^*(c)\rangle_{\psi\otimes\psi}$$

for all $a, b, c \in B$.

Remark 3.1.1. We refer to any functional ψ satisfying the condition in the definition above as a *quantum set functional* on B. The condition $mm^* = \mathrm{Id}_B$ should be thought of as a normalization condition on ψ: not every faithful positive functional on a finite-dimensional C^*-algebra makes (B, ψ) into a quantum set, only those compatible with the multiplication in this precise sense.

Remark 3.1.2. Recall that, more generally, a *comultiplication* on a algebra A is a coassociative linear comultiplication

$$\Delta\colon A \to A \otimes A$$

making the following diagram commute (coassociativity):

$$\begin{tikzcd} A \arrow[r, "\Delta"] \arrow[d, "\Delta"'] & A \otimes A \arrow[d, "\Delta\otimes\mathrm{Id}"] \\ A \otimes A \arrow[r, "\mathrm{Id}\otimes\Delta"'] & A \otimes A \otimes A \end{tikzcd}$$

In general such a Δ is far from unique, and need not interact with any given inner product on A at all. In our setting, however, the situation is much more rigid: once (B, ψ) is a quantum set, the adjoint m^* of the multiplication map is automatically a comultiplication on B, and it is the *unique* one arising this way, since it is pinned down by m together with the Riesz representation theorem for the Hilbert space $L^2(B, \psi)$.

Remark 3.1.3. Because in our setup the comultiplication is obtained from the ordinary multiplication m purely by adjunction with respect to ψ, we will not introduce separate notation for it, and will simply write m^* throughout.

Example 3.1.1 (Classical finite sets as quantum sets)*.* Every finite set V gives rise to a quantum set $(\mathbb{C}^V, \psi_V)$, where $\mathbb{C}^V$ denotes the C^*-algebra of complex-valued functions on V with pointwise multiplication and involution given by pointwise conjugation, $f^*(v) = \overline{f(v)}$, and where

$$\psi_V\colon \mathbb{C}^V \to \mathbb{C}, \qquad \psi_V(f) = \sum_{v\in V} f(v).$$

To see that ψ_V is positive, note that for any $f \in \mathbb{C}^V$ the product f^*f evaluates pointwise as

$$(f^*f)(v) = \overline{f(v)}\, f(v) = |f(v)|^2,$$

then

$$\psi_V(f^*f) = \sum_{v \in V} |f(v)|^2 \geq 0.$$

For faithfulness, suppose $\psi_V(f^*f) = 0$ for some $f \in \mathbb{C}^V$, i.e.

$$\sum_{v \in V} |f(v)|^2 = 0.$$

Since each summand $|f(v)|^2$ is non-negative, the sum can vanish only if every term does, so $f(v) = 0$ for all $v \in V$, and hence $f = 0$. Thus ψ_V is faithful.

It remains to check the quantum set condition $mm^* = \mathrm{Id}$. Let $\{e_v\}_{v \in V}$ denote the orthonormal basis of $\mathbb{C}^V$ given by the characteristic functions of points. Since $e_v e_w = \delta_{vw} e_v$ (pointwise multiplication of indicator functions is nonzero only when $v = w$), the multiplication map satisfies $m(e_v \otimes e_w) = \delta_{vw} e_v$; dualizing this relation with respect to the inner product induced by ψ_V shows that m^* must send each basis vector e_v to the unique diagonal term $e_v \otimes e_v$, i.e.

$$m(f \otimes g) = fg, \qquad m^*(e_v) = e_v \otimes e_v.$$

Evaluating the composite mm^* on a basis element gives

$$(mm^*)(e_v) = m\big(m^*(e_v)\big) = m(e_v \otimes e_v) = e_v e_v = e_v,$$

since e_v is idempotent. As mm^* and Id agree on the basis $\{e_v\}_{v \in V}$, they agree on all of $\mathbb{C}^V$ by linearity, so $mm^* = \mathrm{Id}$, and $(\mathbb{C}^V, \psi_V)$ is indeed a quantum set.

Remark 3.1.4. The identification of finite sets and commutative C^*-algebras is inspired by Gelfand's duality theorem (Negrepontis [13]), which establishes that every commutative C^*-algebra is isomorphic to the algebra of continuous functions on a compact Hausdorff space. In the finite-dimensional case, this space reduces to a discrete finite set, and the algebra's minimal projections correspond directly to the individual elements of the set. Furthermore, the functional ψ_X serves as a quantum analogue of the counting measure (summing over all elements), while the condition $mm^* = \mathrm{Id}$ ensures the correct normalization and structural compatibility of the multiplication map with this measure, generalizing the discrete framework of classical sets to the quantum setting.

Example 3.1.2 (The special case $\mathbb{C}^n$)*.* Taking $V = \{1, \ldots, n\}$ in the previous example recovers the quantum set $(\mathbb{C}^n, \psi_n)$, where ψ_n is determined on the standard basis by $\psi_n(e_i) = 1$ and extends linearly to

$$\psi_n(v) = \sum_{i=1}^{n} v_i, \qquad v \in \mathbb{C}^n.$$

Positivity and faithfulness follow exactly as above: writing $v = \sum_i v_i e_i$ and using orthogonality of the basis,

$$v^*v = \Big(\sum_{i=1}^n \overline{v_i}\, e_i\Big)\Big(\sum_{j=1}^n v_j e_j\Big) = \sum_{i=1}^n \overline{v_i} v_i\, e_i = \sum_{i=1}^n |v_i|^2 e_i,$$

so that

$$\psi_n(v^*v) = \sum_{i=1}^n |v_i|^2 \psi_n(e_i) = \sum_{i=1}^n |v_i|^2 \geq 0,$$

which vanishes precisely when every $v_i = 0$, i.e. when $v = 0$. The quantum set condition again reduces to the computation

$$(mm^*)(e_i) = m(e_i \otimes e_i) = e_i, \qquad i = 1, \dots, n,$$

which forces $mm^* = \mathrm{Id}$ by linearity.

Example 3.1.3 (Matrix algebras). For any $n \in \mathbb{N}_+$, the pair $(M_n(\mathbb{C}), n\,\mathrm{Tr})$ is a quantum set, where Tr denotes the usual trace on $M_n(\mathbb{C})$. We will later see that this quantum set arises naturally as a crossed product: writing $\widehat{\mathbb{Z}}_n$ for the dual of $\mathbb{Z}_n$ acting on $(\mathbb{C}^n, \psi_n)$ via the standard action, one has

$$(M_n(\mathbb{C}), n\,\mathrm{Tr}) \;\cong\; (\mathbb{C}^n \rtimes \widehat{\mathbb{Z}}_n,\ \psi_\rtimes),$$

a fact we prove in 3.1.2. $n\,\mathrm{Tr}$ is chosen instead of simply Tr so that the condition $mm^* = \mathrm{Id}$ is fulfilled.

Lemma 3.1.1 (Comultiplication for matrix algebras)**.** *In the quantum set $(M_n(\mathbb{C}), n\,\mathrm{Tr})$, the comultiplication m^* is given on matrix units by*

$$m^*(e_{ij}) = \frac{1}{n} \sum_{k=1}^n e_{ik} \otimes e_{kj},$$

where e_{ij} denotes the matrix with a single 1 in the i-th row and j-th column, extended linearly to all of $M_n(\mathbb{C})$.

Proof. The proof is a direct computation using two standard facts about matrix units:

1. $e_{ab}e_{cd} = \delta_{bc}\, e_{ad}$;
2. $\langle e_{ij}, e_{kl}\rangle = n\,\mathrm{Tr}(e_{ji}e_{kl}) = n\,\delta_{ik}\,\delta_{jl}$,

where δ_{ij} is the Kronecker delta. In particular, (2) shows that $\{e_{ij}\}_{i,j=1,\dots,n}$ forms an orthogonal (though not orthonormal, since $\langle e_{ij}, e_{ij}\rangle = n$) basis of $M_n(\mathbb{C})$ with respect to the inner product induced by $n\,\mathrm{Tr}$.

Fix arbitrary basis vectors e_{ab}, e_{cd}. By (1),

$$m(e_{ab} \otimes e_{cd}) = e_{ab}e_{cd} = \delta_{bc}\, e_{ad},$$

and hence, for any basis vector e_{ij}, using (2),

$$\big\langle m(e_{ab} \otimes e_{cd}),\, e_{ij} \big\rangle = \delta_{bc} \big\langle e_{ad}, e_{ij} \big\rangle = \delta_{bc}\, n\, \delta_{ai}\, \delta_{dj}.$$

On the other hand, pairing $e_{ab} \otimes e_{cd}$ against the proposed adjoint gives

$$\Big\langle e_{ab} \otimes e_{cd},\, \frac{1}{n} \sum_{k=1}^{n} e_{ik} \otimes e_{kj} \Big\rangle = \frac{1}{n} \sum_{k=1}^{n} \langle e_{ab}, e_{ik} \rangle \langle e_{cd}, e_{kj} \rangle = \frac{1}{n} \sum_{k=1}^{n} \big(n\, \delta_{ai} \delta_{bk} \big) \big(n\, \delta_{ck} \delta_{dj} \big).$$

The sum over k collapses to the single term $k = b = c$, so this equals

$$\frac{1}{n} \cdot n^2\, \delta_{ai}\, \delta_{dj}\, \delta_{bc} = n\, \delta_{ai}\, \delta_{bc}\, \delta_{dj}.$$

The two expressions coincide, so

$$\big\langle m(e_{ab} \otimes e_{cd}), e_{ij} \big\rangle = \big\langle e_{ab} \otimes e_{cd}, m^*(e_{ij}) \big\rangle$$

for all basis vectors, and since $\{e_{ij}\}$ spans $M_n(\mathbb{C})$, this identifies m^* as claimed on the whole of $M_n(\mathbb{C})$ by linearity. $\square$

3.1.1 Crossed product quantum sets

Schäfer and Tobolski [15] introduced the notion of a *crossed product quantum set*. Suppose we are given a finite-dimensional C^*-algebra $\widetilde{B}$ together with a dual group action $\widehat{\alpha}\colon \widehat{G} \to \mathrm{Aut}(\widetilde{B})$, where G is a finite abelian group. By 2.1.1, $G \cong \widehat{G}$. One may then form the crossed product

$$B = \widetilde{B} \rtimes_{\widehat{\alpha}} \widehat{G}.$$

If $\widetilde{B}$ is equipped with a faithful functional $\widetilde{\psi}$ making $(\widetilde{B}, \widetilde{\psi})$ a quantum set, one can construct a corresponding faithful functional on B making B into a quantum set as well, as follows.

Definition 3.1.2 (Crossed product quantum set, Definition 3.1)**.** Let $(\widetilde{B}, \widetilde{\psi})$ be a quantum set, and let $\widehat{\alpha}\colon \widehat{G} \to \mathrm{Aut}(\widetilde{B}, \widetilde{\psi})$ be an action of the dual of a finite abelian group G. The *crossed product quantum set* is defined as

$$(\widetilde{B}, \widetilde{\psi}) \rtimes_{\widehat{\alpha}} \widehat{G} = \big(\widetilde{B} \rtimes_{\widehat{\alpha}} \widehat{G},\ \psi_{\rtimes}\big),$$

where $\widetilde{B} \rtimes_{\widehat{\alpha}} \widehat{G}$ is the crossed product C^*-algebra, and $\psi_{\rtimes}$ is the functional given by

$$\psi_{\rtimes}\Big(\sum_{\chi \in \widehat{G}} b_\chi u_\chi \Big) = n\, \widetilde{\psi}(b_1),$$

with $n = |\widehat{G}|$.

The proof that this definition is well posed, i.e. that the resulting object is indeed a quantum set, can be found in Schäfer and Tobolski [15, Proposition 3.4].

Example 3.1.4 (Example 3.5). If $\widehat{\alpha} = \mathrm{triv}$ is the trivial action, then

$$\widetilde{B} \rtimes_{\widehat{\alpha}} \widehat{G} \;\cong\; \widetilde{B} \otimes C(G),$$

via the isomorphism $b_\chi u_\chi \mapsto b_\chi \otimes \widetilde{u}_\chi$, where $\widetilde{u}_\chi \in C(G)$ denotes the function $g \mapsto \chi(g)$ on G.

Lemma 3.1.2. *Let $\widehat{\alpha}\colon \widehat{\mathbb{Z}}_n \to \mathrm{Aut}(\mathbb{C}^n)$ be the action given by cyclically shifting coordinates. Then*

$$(M_n(\mathbb{C}), n\,\mathrm{Tr}) \;\cong\; (\mathbb{C}^n \rtimes_{\widehat{\alpha}} \widehat{\mathbb{Z}}_n,\ \psi_\rtimes).$$

Proof. Identify $\widehat{\mathbb{Z}}_n$ with $\mathbb{Z}_n = \{0, 1, \ldots, n-1\}$ (arithmetic mod n), and write $\{\delta_i\}_{i \in \mathbb{Z}_n}$ for the standard basis of $\mathbb{C}^n$, so that $\widehat{\alpha}_k(\delta_i) = \delta_{i+k}$. Since $\mathbb{Z}_n$ is finite, every element of $\mathbb{C}^n \rtimes_{\widehat{\alpha}} \widehat{\mathbb{Z}}_n$ is a finite sum $\sum_{k \in \mathbb{Z}_n} b_k u_k$ with $b_k \in \mathbb{C}^n$; writing each $b_k = \sum_i b_k(i)\,\delta_i$ in coordinates, this algebra is spanned, as a vector space, by the n^2 products

$$\delta_i u_k, \qquad i, k \in \mathbb{Z}_n.$$

It therefore suffices to define a $*$-isomorphism $\pi\colon \mathbb{C}^n \rtimes_{\widehat{\alpha}} \widehat{\mathbb{Z}}_n \to M_n(\mathbb{C})(\mathbb{C})$ on this spanning set, check it respects the algebra's product and involution formulas on $\delta_i u_k$, and then extend linearly.

Definition of π. Let $e_{ab} \in M_n(\mathbb{C})(\mathbb{C})$ denote the matrix unit sending $\delta_b \mapsto \delta_a$ and every other $\delta_c \mapsto 0$, and set

$$\pi(\delta_i u_k) = e_{i,\,i-k}, \qquad i, k \in \mathbb{Z}_n,$$

extended linearly to all of $\mathbb{C}^n \rtimes_{\widehat{\alpha}} \widehat{\mathbb{Z}}_n$. Equivalently, writing S for the cyclic shift matrix $S\delta_i = \delta_{i+1}$ and $M_{b_k} = \sum_i b_k(i)\, e_{ii}$ for the diagonal matrix with entries b_k, one checks directly that $\pi(\delta_i u_k) = M_{\delta_i} S^k$, so this agrees with the general formula $\pi\big(\sum_k b_k u_k\big) = \sum_k M_{b_k} S^k$.

π is a $*$-homomorphism. It suffices to check the product and involution formulas on the spanning elements $\delta_i u_k$. For the product, recall that $u_k \delta_i u_k^* = \widehat{\alpha}_k(\delta_i) = \delta_{i+k}$ in $\mathbb{C}^n \rtimes_{\widehat{\alpha}} \widehat{\mathbb{Z}}_n$, matching $S^k M_{\delta_i} S^{-k} = M_{\delta_{i+k}}$ in $M_n(\mathbb{C})(\mathbb{C})$, so

$$\pi(\delta_i u_k)\,\pi(\delta_j u_l) = e_{i,i-k}\, e_{j,j-l} = \delta_{i-k,\,j}\; e_{i,\,j-l},$$

using $e_{ab} e_{cd} = \delta_{bc} e_{ad}$; this is nonzero exactly when $j = i - k$, in which case it equals $e_{i,\,i-k-l} = \pi(\delta_i u_{k+l})$. On the other side,

$$(\delta_i u_k)(\delta_j u_l) = \delta_i\, \widehat{\alpha}_k(\delta_j)\, u_{k+l} = \delta_i\, \delta_{j+k}\, u_{k+l} = \delta_{i,\,j+k}\, \delta_i\, u_{k+l},$$

which is likewise nonzero exactly when $j = i - k$, and then equals $\delta_i u_{k+l}$. Applying π recovers $\pi(\delta_i u_{k+l}) = e_{i,i-k-l}$, matching the computation above. So π respects multiplication on the spanning set, hence on all of $\mathbb{C}^n \rtimes_{\widehat{\alpha}} \widehat{\mathbb{Z}}_n$ by bilinearity.

For the involution, $(\delta_i u_k)^* = \widehat{\alpha}_{-k}(\delta_i^*) u_{-k} = \delta_{i-k}\, u_{-k}$, so $\pi\big((\delta_i u_k)^*\big) = \pi(\delta_{i-k} u_{-k}) = e_{i-k,\,i}$, while directly $\pi(\delta_i u_k)^* = e_{i,i-k}^* = e_{i-k,i}$, matching.

π **is bijective.** As i, k range over $\mathbb{Z}_n$, the pair $(i, i-k)$ ranges exactly once over all of $\mathbb{Z}_n \times \mathbb{Z}_n$ (fix i; varying k makes $i-k$ range over all residues), so $\{\pi(\delta_i u_k)\}_{i,k} = \{e_{ab}\}_{a,b\in\mathbb{Z}_n}$ is exactly the full set of matrix units, a basis of $M_n(\mathbb{C})(\mathbb{C})$. Since $\{\delta_i u_k\}_{i,k}$ is a basis of $\mathbb{C}^n \rtimes_{\widehat{\alpha}} \widehat{\mathbb{Z}}_n$ of the same dimension n^2, and π carries it bijectively to a basis of $M_n(\mathbb{C})(\mathbb{C})$, the linear map π is a bijection.

π **intertwines the functionals.** Since a matrix unit e_{ab} has trace δ_{ab},

$$\operatorname{Tr}\big(\pi(\delta_i u_k)\big) = \operatorname{Tr}(e_{i,i-k}) = \delta_{i,i-k} = \delta_{k,0}.$$

Hence for a general element $b = \sum_k b_k u_k = \sum_{k,i} b_k(i)\,\delta_i u_k$,

$$\operatorname{Tr}(\pi(b)) = \sum_{k,i} b_k(i)\,\delta_{k,0} = \sum_i b_0(i) = \widetilde{\psi}(b_0),$$

where $\widetilde{\psi}(x) = \sum_i x_i$ is the quantum set functional on $(\mathbb{C}^n, \widetilde{\psi})$, and b_0 is exactly the coefficient at the trivial character appearing in the definition of $\psi_\rtimes$. Multiplying by $n = |\widehat{\mathbb{Z}}_n|$,

$$n\operatorname{Tr}(\pi(b)) = n\,\widetilde{\psi}(b_0) = \psi_\rtimes(b).$$

Thus π is a $*$-isomorphism $\mathbb{C}^n \rtimes_{\widehat{\alpha}} \widehat{\mathbb{Z}}_n \to M_n(\mathbb{C})(\mathbb{C})$ satisfying $n\operatorname{Tr}\circ\pi = \psi_\rtimes$, which is exactly an isomorphism of quantum sets

$$(\mathbb{C}^n \rtimes_{\widehat{\alpha}} \widehat{\mathbb{Z}}_n,\ \psi_\rtimes) \ \cong\ (M_n(\mathbb{C})(\mathbb{C}),\ n\operatorname{Tr}). \qquad \square$$

3.2 Quantum graphs

A graph is classically defined as a pair (V, E) with V serving as the *set of vertices* and $E \subseteq V^2$ serving as the *set of edges*. Such a graph could be interpreted as a matrix $A = [a_{ij}] \in \{0,1\}^{n\times n}$ with $(v_i, v_j) \in E \Leftrightarrow a_{ij} = 1$; such a property of such matrices is that they are invariant with respect to representation is often called the *adjacency matrix*, see 3.2.1. The key entrywise multiplication. In turn, we can use this property to generalise graphs on sets to graphs on quantum sets.

Example 3.2.1. Consider the following directed graph on three vertices, featuring a bidirectional connection and a self-loop:

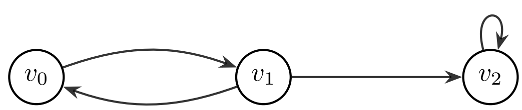


Figure 3.1: A directed graph on three vertices demonstrating varied edge types.

Ordering the basis by $\{v_0, v_1, v_2\}$, the corresponding 3×3 adjacency matrix A is:

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

This linear representation is foundational when generalizing to quantum graphs, where the adjacency matrix $A \in M_n(\mathbb{C})$ is replaced by a linear map between C^*-algebras.

The following definitions are due to Musto, Reutter, and Verdon [12]. We follow the approach of Schäfer and Tobolski [15].

Definition 3.2.1 (Quantum adjacency matrix, Definition 2.5)**.** Let (B, ψ) be a quantum set. A *quantum adjacency matrix* on this quantum set is a linear map $A\colon B \to B$ which is Schur-idempotent and *-preserving, by which we mean that it satisfies

$$m(A \otimes A)m^* = A, \qquad A(b^*) = A(b)^* \quad \text{for all } b \in B.$$

Definition 3.2.2 (Quantum graph, Definition 2.5)**.** A *quantum graph* on (B, ψ) is given by a quantum adjacency matrix on this quantum set.

For all intents and purposes, we will identify both notions and simply call an appropriate $A\colon B \to B$ both a quantum adjacency matrix and a quantum graph.

Remark 3.2.1. There are other equivalent definitions of a quantum graph, namely those based on

- an *edge projection*, i.e. a projection $P \in B \otimes B^{\mathrm{op}}$, see Musto, Reutter, and Verdon [12],
- a $B' - B'$-bimodule $S \subseteq B(\mathcal{H})$, see Weaver [16].

Definition 3.2.3 (Definition 4.6)**.** Let A be a quantum graph on a quantum set (B, ψ). We call the quantum graph

- *loop-free* if $m(A \otimes \mathrm{Id})m^* = 0$,
- *undirected* if $A = A^*$, where the adjoint is taken with respect to the inner product induced by ψ.

Remark 3.2.2. The motivation behind the first point in the definition above will become clear thanks to 3.3.1, which says that $m(A \otimes B)m^*$ can be thought of as taking the intersection of two graphs edgewise. Subsequently $m(A \otimes \mathrm{Id})m^*$ would keep only loops, i.e. edges going from a vertex to itself. The second point is easy to motivate in a similar manner: a graph is undirected if and only if its adjacency matrix is symmetric.

3.3 Voltage and Derived Quantum Graphs

In this section, we introduce the notions of voltage and derived quantum graphs. Voltage graphs and their derived graphs were first introduced by Gross [8]; these notions have since been extended to the setting of quantum graphs by Schäfer and Tobolski [15].

3.3.1 Classical voltage graphs

Before turning to the quantum setting, it is instructive to first recall the classical construction. Informally, a voltage graph is simply a graph whose edges have been labeled, or *weighted*, by elements of a finite group; the derived graph then unfolds this labeled graph into a (typically much larger) covering graph, built by tracking how these group elements accumulate as one walks along paths in the original graph. This construction originates in topological graph theory [7], where it was used to build graph embeddings on surfaces of higher genus, but it also turns out to provide a rich and flexible source of examples of graphs with interesting symmetry, which is the perspective we adopt here.

Definition 3.3.1 (Multigraph, Diestel [3], Section 1.10)**.** A *multigraph* $\Gamma = (V, E)$ is a graph in which multiple distinct edges may join the same pair of vertices, i.e. E is a multiset on $V \times V$ rather than a subset.

Definition 3.3.2 (Classical voltage graph, Gross and Tucker [7], Section 2.1)**.** A *voltage graph* is a pair (Γ, λ) consisting of a finite directed multigraph Γ, with vertex set V and edge set E, together with a labelling $\lambda\colon E \to G$ of the edges of Γ by elements of a finite group G.

From every voltage graph one can construct a new graph, called its *derived graph*, as follows.

Definition 3.3.3 (Classical derived graph, Gross and Tucker [7], Section 2.1)**.** The *derived graph* Γ^λ of (Γ, λ) has vertex set $V \times G$ and edge set $E \times G$. Whenever $e \in E$ is an edge from u to v in Γ, and $g \in G$ is a group element, the corresponding edge (e, g) in Γ^λ runs from (u, g) to $(v,\, g\lambda(e))$.

Intuitively, the derived graph consists of $|G|$ labeled copies (*fibers*) of the vertex set of Γ, one for each group element, and every edge e of Γ is lifted to $|G|$ edges in Γ^λ, one leaving each fiber, whose endpoint is determined by translating within the target fiber according to the voltage $\lambda(e)$. This is precisely the picture underlying the examples given below.

The following is a major result in the theory of voltage and derived graphs, showing that this construction is not merely a convenient way of producing examples, but in fact accounts for *every* graph admitting a free action of a finite group.

Definition 3.3.4 (Graph automorphism, Diestel [3], Section 1.1)**.** Let $\Gamma = (V, E)$ be a graph. An *automorphism* of Γ is a bijection $f\colon V \to V$ such that

$$(u, v) \in E \iff (f(u), f(v)) \in E$$

for all $u, v \in V$, i.e. a permutation of the vertex set that preserves adjacency. The set of all automorphisms of Γ, equipped with composition, forms a group, denoted $\mathrm{Aut}(\Gamma)$.

Definition 3.3.5 (Group action on a graph)**.** Let $\Gamma = (V, E)$ be a graph and let G be a group. A *group action* of G on Γ is a group homomorphism

$$\alpha \colon G \to \mathrm{Aut}(\Gamma).$$

As with group actions on C^*-algebras, we write $g \cdot v$ instead of $\alpha(g)(v)$ when the action in question is clear. The action is called *free* if $g \cdot v = v$ for some $v \in V$ implies $g = e_G$.

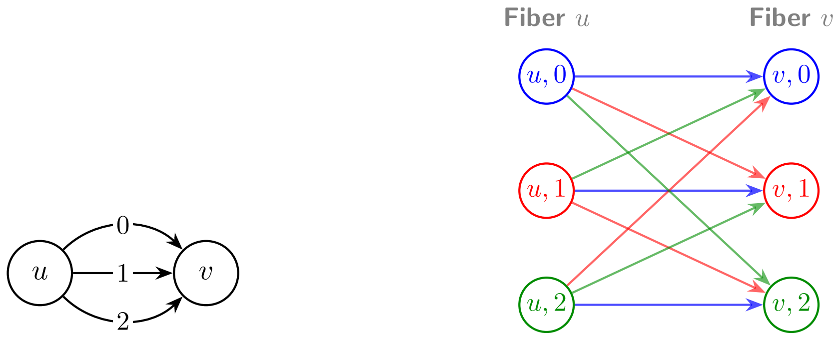
Definition 3.3.6 (Quotient graph)**.** Let $\Gamma = (V, E)$ be a graph and let G act on Γ. The *quotient graph* Γ/G is the graph with vertex set V/G, the set of orbits of G on V, and with an edge from the orbit $[u]$ to the orbit $[v]$ whenever $(u', v') \in E$ for some $u' \in [u]$, $v' \in [v]$.

Theorem 3.3.1 (Gross and Tucker theorem, Gross and Tucker [7], Theorem 2.2.2)**.** *Let $\widetilde{\Gamma}$ be a graph and let G be a finite group acting freely on $\widetilde{\Gamma}$ by automorphisms. Then there exists a voltage graph (Γ, λ), with edges labeled by elements of G, such that $\widetilde{\Gamma}$ is isomorphic to the derived graph of (Γ, λ). The graph Γ is the quotient graph $\widetilde{\Gamma}/G$.*

In other words, voltage graphs classify free group actions on graphs up to isomorphism: recovering Γ from $\widetilde{\Gamma}$ amounts to nothing more than quotienting by the group action. However, this is not enough to recover the voltage labelling λ, as different labellings might create isomorphic derived graphs.

Base voltage graph Γ over $\mathbb{Z}_3$ Derived graph Γ^λ (two 3-cycles)

Figure 3.2: Voltage graph and derived cover over $\mathbb{Z}_3$, for a base graph with a single vertex and two loops.



Base theta graph Γ with edge labels Derived graph Γ^λ

Figure 3.3: Voltage graph and derived cover for a theta graph over $\mathbb{Z}_3$.

3.3.2 Generalisation to Voltage Quantum and Derived Quantum Graphs

We now turn to generalising the Gross–Tucker construction to the setting of quantum graphs. This generalisation was first carried out by Schäfer and Tobolski [15], building on the notion of a crossed product quantum set introduced earlier in this chapter. Just as a classical voltage graph is a labelling of the edges of a graph by group elements, a voltage *quantum* graph will be a family of quantum adjacency matrices indexed by the group G rather than a single labelling function; the derived quantum graph then reassembles this family into a single quantum adjacency matrix on the crossed product quantum set $(B, \psi_\rtimes)$, playing the role of the derived graph Γ^λ from the classical picture. The following definition makes this precise.

Definition 3.3.7 (Schäfer and Tobolski [15], Definition 4.1)**.** Suppose $(\widetilde{B}, \widetilde{\psi})$, the action $\widehat{\alpha}\colon \widehat{G} \to \mathrm{Aut}(\widetilde{B}, \widetilde{\psi})$, the crossed product $(B, \psi_\rtimes)$, and the elements $(u_\chi)_{\chi\in\widehat{G}} \subseteq B$ are as introduced in 2.2.1.

1. For crossed product quantum set $\widetilde{B} \rtimes_{\widehat{\alpha}} \widehat{G}$ we define the linear map

$$E_\rtimes : \widetilde{B} \rtimes_{\widehat{\alpha}} \widehat{G} \to \widetilde{B}, \qquad \sum_{\chi\in\widehat{G}} b_\chi u_\chi \mapsto n b_1,$$

   and my $E_\rtimes^*$ its adjoint with respect to the inner products on $\widetilde{B}$ and $\widetilde{B} \rtimes_{\widehat{\alpha}} \widehat{G}$.

2. For each $g \in G$, set

$$X_g = \frac{1}{n} \sum_{\chi\in\widehat{G}} \overline{\chi(g)}\, u_\chi u_\chi^*,$$

   i.e. let X_g be the linear operator

$$X_g\colon B \to B, \qquad b \mapsto \frac{1}{n} \sum_{\chi\in\widehat{G}} \overline{\chi(g)}\, \langle u_\chi, b\rangle_{\psi_\rtimes} u_\chi.$$

3. Given the group action $\widehat{\alpha}\colon \widehat{G} \to \mathrm{Aut}(\widetilde{B}, \widetilde{\psi})$, a *voltage quantum graph* on $(\widetilde{B}, \widetilde{\psi})$ is a G-indexed family $\widetilde{A} = (\widetilde{A}_g)_{g \in G}$ of quantum adjacency matrices such that
$$\widetilde{A}_g \widehat{\alpha}_\chi = \widehat{\alpha}_\chi \widetilde{A}_g$$
holds for every $g \in G$ and every $\chi \in \widehat{G}$.

4. Given such a family $\widetilde{A}$, the associated *derived quantum graph* $\widetilde{A} \rtimes_{\widehat{\alpha}} \widehat{G}\colon B \to B$ on $(B, \psi_\rtimes)$ is
$$\widetilde{A} \rtimes_{\widehat{\alpha}} \widehat{G} = \sum_{g \in G} m\big(X_g \otimes E_\rtimes^* \widetilde{A}_g E_\rtimes\big)\, m^*,$$
where $m = m_\rtimes$ is the multiplication map on B.

The role played by X_g here is directly analogous to the indicator function of the fiber $V \times \{g\}$ in the classical derived graph: it isolates, within B, the *copy of* g sitting inside the crossed product. The fact that X_g is itself an adjacency matrix on the quantum set $(B, \psi_\rtimes)$ is checked in Schäfer and Tobolski [15, Proposition 4.5], and the derived quantum graph is then built by combining these fiber projections with the individual voltage components $\widetilde{A}_g$, summed over all $g \in G$, mirroring how, in the classical construction, each edge of Γ^λ is assigned both to a specific fiber and to a specific lift of an edge of Γ.

As in the classical case, it will be useful to know when a voltage quantum graph gives rise to a derived quantum graph with no loops, or one that is undirected.

Definition 3.3.8. A voltage quantum graph $\widetilde{A} = \{\widetilde{A}_g\}_{g \in G}$ is called *loop-free* if $\widetilde{A}_1$ is loopfree in the sense of 3.2.3. $\widetilde{A}$ is called *undirected* if $\widetilde{A}_g^* = \widetilde{A}_{g^{-1}}$ holds for all $g \in G$.

We can connect the notions of voltage and derived graphs to their quantum equivalents. The following proposition tells us that classical voltage graphs can be thought of as voltage quantum graphs. Below $(\mathbb{C}^V, \psi_V)$ is used to denote a quantum set derived from a finite classical set V, like it was done in Example 3.1.1.

Definition 3.3.9 (Pre-simple voltage graph)**.** A voltage graph (Γ, λ) is called *pre-simple* if its derived graph Γ^λ is simple.

Proposition 3.3.1 (Schäfer and Tobolski [15], Proposition 4.10)**.** *Let G be a finite abelian group, and let $(\mathbb{C}^V, \psi_V)$ be a quantum set. Given a pre-simple voltage graph (Γ, λ) on V with labelling $\lambda\colon E \to G$, let $\widetilde{A}_g\colon \mathbb{C}^V \to \mathbb{C}^V$ denote the adjacency matrix of the (simple, directed) subgraph of Γ consisting of exactly those edges labeled g. Then the assignment*
$$(\Gamma, \lambda) \longmapsto \widetilde{A} = (\widetilde{A}_g)_{g \in G}$$
is a one-to-one correspondence between G-labeled pre-simple voltage graphs on V and voltage quantum graphs on $(\mathbb{C}^V, \psi_V)$ with respect to the trivial action on $(\mathbb{C}^V, \psi_V)$.

In the same manner certain classical derived graphs can be identified with derived quantum graphs. Note that the group action in question is trivial, which connects to Theorem 3.3.1, which states that a graph is isomorphic to a derived graph of some voltage graph in and only if it allows a free group action of some fixed finite group.

Proposition 3.3.2 (Schäfer and Tobolski [15], Proposition 4.12)**.** *In the setting of Proposition 3.3.1, let* (Γ, λ) *be a pre-simple voltage graph with associated voltage quantum graph* $\widetilde{A} = (\widetilde{A}_g)_{g\in G}$. *Then the derived graphs agree under the identification*

$$\Gamma^\lambda = \widetilde{A} \rtimes_{\mathrm{triv}} \widehat{G},$$

where Γ^λ *is the Gross–Tucker derived graph, and* $\widetilde{A} \rtimes_{\mathrm{triv}} \widehat{G}$ *is the derived quantum graph of Definition 3.3.7; here classical graphs on* $V \times G$ *are identified with quantum graphs on* $(\mathbb{C}^V, \psi_V) \rtimes_{\mathrm{triv}} \widehat{G}$.

Thus we can identify classical derived graphs with quantum graphs on cross-product quantum sets with the trivial action. This will come useful later as this will allow us to establish a quantum isomorphism between nonclassical graphs on $(M_3(\mathbb{C}), 3\,\mathrm{Tr})$ and classical graphs that happen to be derived graphs.

The lemma below lets us better understand the correspondence between classical and quantum derived graphs.

Lemma 3.3.1 (Schäfer and Tobolski [15], Lemma 4.11)**.** *Let* $(\mathbb{C}^V, \psi_V)$ *be the quantum set associated with a finite set* V*, and let* G *be a finite abelian group equipped with the trivial action* $\mathrm{triv}\colon \widehat{G} \to \mathrm{Aut}(\mathbb{C}^V, \psi_V)$. *Then following hold.*

1. *The assignment on generators*

$$e_v \mapsto \sum_{g\in G} e_{v,g}, \qquad u_\chi \mapsto \sum_{v\in V,\, g\in G} \chi(g)\, e_{v,g}$$

extends to an isomorphism

$$(\mathbb{C}^V, \psi_V) \rtimes_{\mathrm{triv}} \widehat{G} \;\cong\; (\mathbb{C}^{V\times G}, \psi_{V\times G}),$$

where the right-hand side is the quantum set associated to the finite set $V \times G$ *as in Example 3.1.1, and* e_v, $e_{v,g}$ *denote the indicator functions in* $\mathbb{C}^V$ *and* $C(V \times G)$*, respectively.*

2. *Under this identification, the operator* $X_g\colon B \to B$, $g \in G$, *from Definition 3.3.7 corresponds to the classical graph on* $V \times G$ *whose edges are exactly*

$$(v, h) \to (w, hg), \qquad v, w \in V,\ h \in G.$$

3. *Likewise, the operator* $E_\rtimes^* \widetilde{A}_g E_\rtimes \colon B \to B$ *from Definition 3.3.7 corresponds to the classical graph on* $V \times G$ *whose edges are exactly*

$$(v, h) \to (w, k), \qquad h, k \in G,\ v \xrightarrow{\widetilde{A}_g} w,$$

where $v \xrightarrow{\widetilde{A}_g} w$ *means that* v *and* w *are joined by an edge in the classical graph corresponding to the adjacency matrix* $\widetilde{A}_g$.

4. *Finally, suppose* A_1, A_2 *are quantum adjacency matrices corresponding, in the sense of Example 3.1.1, to classical graphs* Γ_1, Γ_2 *on* $V \times G$. *Then*

$$A_3 = m(A_1 \otimes A_2)\, m^*$$

is again a quantum adjacency matrix, and corresponds to the classical graph $\Gamma_3 = \Gamma_1 \cap \Gamma_2$ *with vertex set* $V \times G$ *and edge set given by the intersection of the edge sets of* Γ_1 *and* Γ_2.

Example 3.3.1 (Schäfer and Tobolski [15], Example 4.13). Consider the voltage graph (Γ, λ) consisting of one vertex v along with a single edge e going from v to itself and labeled with $1_{\mathbb{Z}_3}$, i.e. $\lambda(e) = 1_{\mathbb{Z}_3}$. The derived graph then has has three, one for each element of $\mathbb{Z}_3$.

Base voltage graph Γ Derived graph Γ^λ

Figure 3.4: Base voltage graph and its derived cover over $\mathbb{Z}_3$.

As discussed above, a classical graph on n vertices corresponds to a quantum graph on the quantum set $(\mathbb{C}^n, \psi_n)$. The voltage graph (Γ, λ) is thus identified with a voltage quantum graph on $(\mathbb{C}, \psi_1)$, given by a family $(\widetilde{A}_g)_{g \in \mathbb{Z}_3}$ of quantum adjacency matrices on $(\mathbb{C}, \psi_1)$. Since Γ has only a single edge, labeled by $1_{\mathbb{Z}_3}$, it is straightforward to check that (Γ, λ) corresponds to the family $\widetilde{A} = (\widetilde{A}_g)_{g \in \mathbb{Z}_3}$ given by

$$\widetilde{A}_{0_{\mathbb{Z}_3}} = 0, \qquad \widetilde{A}_{1_{\mathbb{Z}_3}} = \mathrm{id}_{\mathbb{C}}, \qquad \widetilde{A}_{2_{\mathbb{Z}_3}} = 0.$$

In general one would need to verify that $(\widetilde{A}_g)_{g \in \mathbb{Z}_3}$ satisfies the conditions of Definition 3.3.7 in order to qualify as a voltage quantum graph; here, however, we are regarding $(\widetilde{A}_g)_{g \in \mathbb{Z}_3}$ as a voltage quantum graph with respect to the trivial action triv of $\mathbb{Z}_3$ on $\mathbb{C}$, for which there is nothing left to check (see Proposition 3.3.1).

What is the derived quantum graph $\widetilde{A} \rtimes_{\mathrm{triv}} \mathbb{Z}_3$ of this voltage quantum graph? Recall first that the derived quantum graph lives on the crossed product quantum set

$$(\mathbb{C}, \psi_1) \rtimes_{\mathrm{triv}} \mathbb{Z}_3.$$

By part (a) of Lemma 3.3.1, this quantum set is isomorphic to $(\mathbb{C}^3, \psi_3)$, and quantum graphs on $(\mathbb{C}^3, \psi_3)$ correspond to classical graphs on three vertices.

By Definition 3.3.7, the derived quantum graph $A = \widetilde{A} \rtimes_{\mathrm{triv}} \mathbb{Z}_3 \colon \mathbb{C}^3 \to \mathbb{C}^3$ on $(\mathbb{C}^3, \psi_3)$ is given by

$$A = \sum_{g \in \mathbb{Z}_3} m\big(X_g \otimes E_{\rtimes}^{*} \widetilde{A}_g E_{\rtimes}\big)\, m^{*}.$$

By parts (b) and (c) of Lemma 3.3.1, the quantum graphs X_g and $E_{\rtimes}^{*} \widetilde{A}_g E_{\rtimes}$ correspond to the classical graphs on $\{v_0, v_1, v_2\}$ shown in Figure 3.5 and Figure 3.6, respectively.

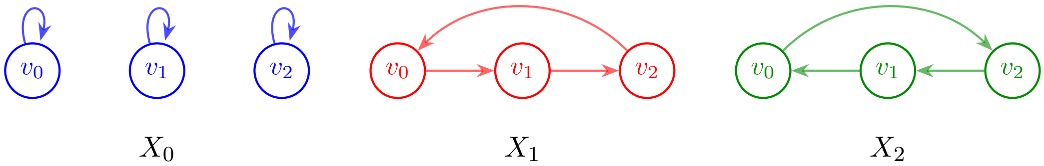


Figure 3.5: The graphs corresponding to the adjacency matrices X_g.

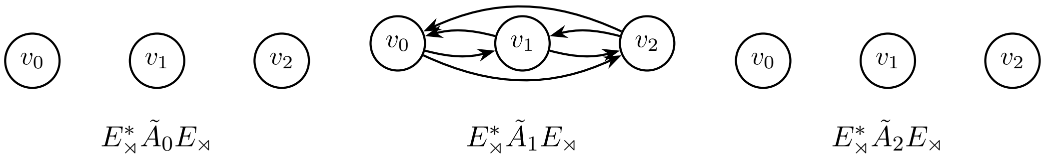


Figure 3.6: The graphs corresponding to the adjacency matrices $E_{\rtimes}^{*} \tilde{A}_g E_{\rtimes}$.

By part (d) of Lemma 3.3.1, the term

$$m\big(X_g \otimes E_{\rtimes}^{*} \widetilde{A}_g E_{\rtimes}\big)\, m^{*}$$

returns the adjacency matrix of the classical graph obtained by intersecting the graphs of Figure 3.5 and Figure 3.6. Only the term for $g = 1_{\mathbb{Z}_3}$ contributes to the derived quantum graph A, so that

$$A = m\big(X_1 \otimes E_{\rtimes}^{*} \widetilde{A}_1 E_{\rtimes}\big)\, m^{*},$$

and it is straightforward to check that this is exactly the derived graph Γ^{λ} from Figure 3.4.

This confirms, in the simplest possible case, the general correspondence of Proposition 3.3.1 and Proposition 3.3.2: starting from a classical voltage graph, treating it as a voltage quantum graph with respect to the trivial action reproduces exactly the classical derived graph via Definition 3.3.7.

3.4 Isomorphisms and Quantum Isomorphisms of Quantum Sets

Quantum isomorphisms were first conceived in Atserias et al. [1] in the context of game theory. We follow the definitions due to Schäfer and Tobolski [15], which in turn build on the work of Musto, Reutter, and Verdon [12].

Definition 3.4.1 (Quantum isomorphisms, Schäfer and Tobolski [15], Definition 2.8)**.** Let (B_1, ψ_1) and (B_2, ψ_2) be quantum sets, and let $A_1 \colon B_1 \to B_1$ and $A_2 \colon B_2 \to B_2$ be quantum graphs on (B_1, ψ_1) and (B_2, ψ_2), respectively.

- The quantum sets (B_1, ψ_1) and (B_2, ψ_2) are called *isomorphic* if there exists a $*$-isomorphism $\phi \colon B_1 \to B_2$ satisfying $\psi_2 \circ \phi = \psi_1$. The group of all such self-isomorphisms of a quantum set (B, ψ) is denoted $\mathrm{Aut}(B, \psi)$.
- Two quantum graphs are called *isomorphic* if the underlying quantum sets admit an isomorphism $\phi \colon B_1 \to B_2$ which additionally intertwines the graphs, i.e. satisfies $\phi A_1 = A_2 \phi$. The corresponding group of automorphisms of a quantum graph A is written $\mathrm{Aut}(A)$.
- The quantum sets (B_1, ψ_1) and (B_2, ψ_2) are said to be *quantum isomorphic*, denoted $(B_1, \psi_1) \cong_q (B_2, \psi_2)$, if there exist a finite-dimensional Hilbert space $\mathcal{H}$ together with a unital $*$-homomorphism

  $$\rho \colon B_1 \to B_2 \otimes B(\mathcal{H})$$

  for which the induced map

  $$p \colon L^2(B_1) \otimes \mathcal{H} \to L^2(B_2) \otimes \mathcal{H}, \qquad a \otimes \xi \longmapsto \rho(a)(1 \otimes \xi),$$

  is a unitary operator between Hilbert spaces.
- Two quantum graphs A_1, A_2 are *quantum isomorphic*, written $A_1 \cong_q A_2$, if the underlying quantum sets admit a quantum isomorphism $\rho \colon B_1 \to B_2 \otimes B(\mathcal{H})$ that in addition satisfies the intertwining condition

  $$\rho A_1 = (A_2 \otimes \mathrm{Id}_{B(\mathcal{H})})\, \rho.$$

Remark 3.4.1. Taking $\mathcal{H} = \mathbb{C}$ in the definition above recovers the ordinary notion of isomorphism as a special case: a unital $*$-homomorphism $\rho \colon B_1 \to B_2 \otimes B(\mathbb{C}) = B_2$ for which $p = \rho$ is unitary is exactly a $*$-isomorphism $\phi \colon B_1 \to B_2$ satisfying $\psi_2 \circ \phi = \psi_1$, and the intertwining condition $\rho A_1 = (A_2 \otimes \mathrm{Id})\rho$ reduces to $\phi A_1 = A_2 \phi$. In this sense, quantum isomorphisms genuinely generalize the classical notion above.

Quantum isomorphism is thus a strictly weaker notion than ordinary isomorphism: every isomorphism of quantum sets or quantum graphs is in particular a quantum isomorphism, but not conversely. An example is given in 3.4.1. Later, we will use this weaker notion to establish quantum isomorphisms between certain classical derived graphs and quantum graphs on $(M_3(\mathbb{C}), 3\,\mathrm{Tr})$ that are not, and cannot be, isomorphic in the ordinary sense.

Proposition 3.4.1. *Quantum isomorphism $\cong_q$ is strictly weaker than ordinary isomorphism $\cong$ for quantum sets and quantum graphs.*

Proof. Consider the matrix algebra $B_1 = M_2(\mathbb{C})$ and the commutative algebra $B_2 = \mathbb{C}^4$, equipped with the quantum set functionals established earlier in this thesis:

$$\psi_1(x) = 2\operatorname{Tr}(x), \qquad \psi_2(y) = \sum_{k=0}^{3} y_k.$$

Classical non-isomorphism. The C^*-algebras B_1 and B_2 cannot be classically isomorphic, since $B_1 = M_2(\mathbb{C})$ is non-commutative while $B_2 = \mathbb{C}^4$ is commutative, and any $*$-isomorphism would have to preserve commutativity. Thus $(B_1, \psi_1) \ncong (B_2, \psi_2)$.

Existence of a quantum isomorphism. Set $\mathcal{H} = \mathbb{C}^2$. Let $\{e_0, e_1, e_2, e_3\}$ denote the standard basis of minimal projections of $\mathbb{C}^4$, and let $\sigma_0 = I_2$ together with the Pauli matrices $\sigma_1, \sigma_2, \sigma_3$ of Example 2.3.2 form the family $\{\sigma_0, \sigma_1, \sigma_2, \sigma_3\} \subseteq U(2)$.

Define the unital $*$-homomorphism $\rho \colon M_2(\mathbb{C}) \to \mathbb{C}^4 \otimes M_2(\mathbb{C})$ by

$$\rho(x) = \sum_{k=0}^{3} e_k \otimes \sigma_k x \sigma_k^*.$$

The induced map $p \colon L^2(M_2(\mathbb{C})) \otimes \mathbb{C}^2 \to L^2(\mathbb{C}^4) \otimes \mathbb{C}^2$ acts as $p(a \otimes \xi) = \rho(a)(1 \otimes \xi)$. Since the e_k are pairwise orthogonal with $\psi_2(e_k) = 1$, and $\sigma_k^* \sigma_k = I_2$, a direct computation gives

$$\begin{aligned}
\langle p(a \otimes \xi), p(b \otimes \eta) \rangle &= \sum_{k=0}^{3} \operatorname{Tr}\left(\xi^* \sigma_k (a^* b) \sigma_k^* \eta\right) \\
&= \operatorname{Tr}\left(\xi^* \left(\sum_{k=0}^{3} \sigma_k (a^* b) \sigma_k^*\right) \eta\right).
\end{aligned}$$

By the Pauli twirl identity $\sum_{k=0}^{3} \sigma_k Y \sigma_k^* = 2\operatorname{Tr}(Y) I_2$, valid for any $Y \in M_2(\mathbb{C})$, this simplifies to

$$\langle p(a \otimes \xi), p(b \otimes \eta) \rangle = 2\operatorname{Tr}(a^* b)\, \langle \xi, \eta \rangle_{\mathbb{C}^2} = \psi_1(a^* b)\, \langle \xi, \eta \rangle_{\mathbb{C}^2} = \langle a, b \rangle_{L^2(M_2(\mathbb{C}))}\, \langle \xi, \eta \rangle_{\mathbb{C}^2}.$$

Thus p preserves inner products. Since $\dim(L^2(B_1) \otimes \mathcal{H}) = 4 \times 2 = 8 = \dim(L^2(B_2) \otimes \mathcal{H})$, p is a unitary operator.

Taking the quantum graphs $A_1 = 0$ on (B_1, ψ_1) and $A_2 = 0$ on (B_2, ψ_2), the intertwining condition $\rho A_1 = (A_2 \otimes \operatorname{Id})\rho$ holds trivially, since both sides vanish.

Hence $A_1 \cong_q A_2$, while $(B_1, \psi_1) \ncong (B_2, \psi_2)$ rules out $A_1 \cong A_2$ as well. This proves the claim. ☐

The theorem below lets us establish quantum isomorphisms between derived quantum graphs differing only by the choice of group action.

Theorem 3.4.1 (Schäfer and Tobolski [15], Theorem 5.2)**.** *Let $(\widetilde{B}, \widetilde{\psi})$, the action $\widehat{\alpha} \colon \widehat{G} \to \operatorname{Aut}(\widetilde{B}, \widetilde{\psi})$, and the crossed product $(\widetilde{B} \rtimes_{\widehat{\alpha}} \widehat{G}, \psi_{\rtimes})$ be as introduced above,* *and let $\widetilde{A} = (\widetilde{A}_g)_{g\in G}$ be a voltage quantum graph on $(\widetilde{B}, \widetilde{\psi})$ with respect to the action $\widehat{\alpha}$ of $\widehat{G}$. Then the corresponding derived quantum graphs satisfy the quantum isomorphism*

$$\widetilde{A} \rtimes_{\widehat{\alpha}} \widehat{G} \cong_q \widetilde{A} \rtimes_{\mathrm{triv}} \widehat{G}.$$

What is truly remarkable in 3.4.1 is that we can start with a classical pre-simple voltage graph with dual group action that is compatible with the labeling and obtain a genuine quantum graph, i.e. one that is not isomorphic to any classical graph.

Theorem 3.4.1 will be central to our strategy in Section 4.3 by choosing $\widehat{\alpha}$ to be the trivial action, we already know from Proposition 3.3.1 and Lemma 3.3.1 that $\widetilde{A} \rtimes_{\mathrm{triv}} \widehat{G}$ is (isomorphic to) a *classical* derived graph. The above theorem then lets us transport this classical realization back along any other action $\widehat{\alpha}$ we might prefer to work with, establishing a quantum isomorphism between the (possibly non-classical) derived quantum graph $\widetilde{A} \rtimes_{\widehat{\alpha}} \widehat{G}$ and a genuinely classical graph, without having to construct that isomorphism by hand in each individual case. This is precisely the tool we will use to show that every one of the five equivalence classes of loop-free, undirected quantum graphs on $(M_3(\mathbb{C}), 3\,\mathrm{Tr})$ established in Corollary **??** is quantum isomorphic to a classical graph on nine vertices.

3.5 Gromada's theorem

In this section, we state a theorem due to Gromada [6] establishing a correspondence between certain linear subspaces of $\mathfrak{sl}(n)$ and quantum graphs. This result will be used to classify all loop-free and undirected quantum graphs on $M_3(\mathbb{C})$ up to quantum isomorphism, and it also provides an explicit recipe for recovering the adjacency matrix corresponding to a given subspace.

Definition 3.5.1 (Quantum Edges)**.** Let $A_V : M_n(\mathbb{C}) \to M_n(\mathbb{C})$ be a quantum graph on $M_n(\mathbb{C})$ associated with a subspace $V \subseteq \mathfrak{gl}(n, \mathbb{C})$. The *number of quantum edges* of the quantum graph A_V is defined as the dimension of V:

$$\text{q-edges}(A_V) = \dim(V).$$

Theorem 3.5.1 (Gromada [6], Theorem 3.7)**.** *Every linear map $A : M_n(\mathbb{C}) \to M_n(\mathbb{C})$ defining a quantum graph on $M_n(\mathbb{C})$ corresponds uniquely to a subspace $V \subseteq \mathfrak{gl}(n, \mathbb{C})$ via*

$$A = n^2 \sum_{s=1}^{\dim(V)} \xi_s \otimes \bar{\xi}_s,$$

where $(\xi_1, \ldots, \xi_{\dim(V)})$ is any orthonormal basis of V with respect to the Hilbert–Schmidt inner product.

Furthermore, the quantum graph A:

1. *Has $m = \dim(V)$ quantum edges;*

2. *Has no loops if and only if* $V \subseteq \mathfrak{sl}(n)$*;*

3. *Is undirected if and only if* $V = V^* = \{x^* \mid x \in V\}$*.*

If two subspaces $V, W \subseteq \mathfrak{gl}(n, \mathbb{C})$ *determine isomorphic quantum graphs* $A_V \cong A_W$*, then* $V = UWU^*$ *for some unitary matrix* $U \in SU(n)$*.*

The subspace corresponding to a quantum graph by 3.5.1 will be called its *Gromada subspace.*

Remark 3.5.1. The statement of Theorem 3.5.1 differs from its original formulation by a factor of n^2 in front of the sum. This rescaling is necessary to account for the different conventions used in the two works.

Before applying this correspondence, it will be convenient to have a more concrete handle on subspaces satisfying $V = V^*$, since this is precisely the condition singling out undirected quantum graphs. The next two lemmas establish that such subspaces always admit a basis of self-adjoint elements, and that this self-adjoint data is in fact enough to reconstruct V uniquely.

Lemma 3.5.1 (Gromada [6], Lemma 3.8)**.** *Every vector subspace* $V \subseteq \mathfrak{gl}(n, \mathbb{C})$ *with* $V = V^*$ *admits a self-adjoint basis* $\{\xi_s\}$*, i.e. with* $\xi_s = \xi_s^*$ *for every* s*.*

Proof. We construct the desired basis inductively. Suppose we have already obtained a linearly independent set of self-adjoint vectors $\{\xi_s\}$ spanning a subspace $V_1 \subseteq V$. By construction, V_1 is invariant under the involution, i.e. $V_1 = V_1^*$.

If $V_1 \neq V$, let $V_2 = V_1^\perp$ be the orthogonal complement of V_1 with respect to the standard inner product. Since the inner product and the involution are compatible, V_2 is also invariant under the adjoint. Choose any nonzero $\xi \in V_2$ and decompose it into its self-adjoint real and imaginary parts,

$$\xi_{\mathrm{Re}} = \frac{\xi + \xi^*}{2}, \qquad \xi_{\mathrm{Im}} = \frac{i(\xi^* - \xi)}{2}.$$

Both ξ_{Re} and ξ_{Im} are self-adjoint elements of V_2, and since $\xi \neq 0$, at least one of them is nonzero. We extend our independent set by appending either both (if linearly independent) or the single nonzero one (if not). Iterating this process eventually produces a full self-adjoint basis of V. $\square$

The following lemma shows that any undirected quantum graph on $M_n(\mathbb{C})$ may be identified with a real vector space.

Lemma 3.5.2 (Gromada [6], Remark 3.10)**.** *A complex vector space* $V = V^* \subseteq \mathfrak{gl}(n, \mathbb{C})$ *uniquely determines a real vector space, spanned by a basis* $\{\xi_s\}$ *of self-adjoint elements* $\xi_s = \xi_s^*$*.*

Proof. Let $V \subseteq \mathfrak{gl}(n, \mathbb{C})$ be a vector space with $V = V^*$.

Since V is invariant under the Hermitian adjoint, every $v \in V$ admits the decomposition

$$v = \frac{v + v^*}{2} + i\,\frac{v - v^*}{2i}.$$

Both summands lie in V, as V is closed under addition, scalar multiplication, and the adjoint operation, and moreover

$$\left(\frac{v + v^*}{2}\right)^* = \frac{v + v^*}{2}, \qquad \left(i\,\frac{v - v^*}{2i}\right)^* = i\,\frac{v - v^*}{2i},$$

so both are self-adjoint.

Now let $\xi \in V$ be self-adjoint, and let $\{\xi_s\}$ be a self-adjoint basis of V. Write $\xi = \sum_s c_s \xi_s$ for some $c_s \in \mathbb{C}$. Taking the Hermitian adjoint gives

$$\xi = \xi^* = \sum_s \overline{c_s}\,\xi_s,$$

since each ξ_s is self-adjoint, and by uniqueness of the expansion with respect to $\{\xi_s\}$ we get $c_s = \overline{c_s}$ for every s, i.e. every coefficient c_s is real.

Hence every self-adjoint element of V is a real linear combination of $\{\xi_s\}$, so the self-adjoint part of V is a real vector space uniquely determined by V. □

By the above lemma, we may regard any self-adjoint $V \subseteq \mathfrak{sl}(n)$ as a real subspace of $\mathfrak{sl}(n)$.

Remark 3.5.2 (Gromada [6], Section 2.3). For a quantum set (B, ψ), it is tempting to call the minimal central projections of B (i.e. the matrix blocks M_{n_i} in a decomposition $B \cong \bigoplus_i M_{n_i}$) its *vertices*. This works well enough for classical sets, where $B = \mathbb{C}^n$ decomposes into n copies of $M_1(\mathbb{C}) = \mathbb{C}$, recovering the usual n vertices. However, for a genuinely noncommutative quantum set such as $(M_n(\mathbb{C}), n\,\mathrm{Tr})$, this convention is misleading: it would count $M_n(\mathbb{C})$ as a single *vertex*, even though $\dim(M_n(\mathbb{C})) = n^2$ and the quantum graphs living on $M_n(\mathbb{C})$ behave, in every respect relevant to us, like graphs on n^2 points rather than on a single point.

This is exactly why we define q-edges(A_V) as the dimension $\dim(V)$ of the corresponding subspace $V \subseteq \mathfrak{gl}(n, \mathbb{C})$, rather than by some more naive combinatorial count: it is the dimension of the underlying vector space, not a literal count of *vertices* or *edges* in any classical sense, that yields consistent statements such as Corollary 4.2.1 and the classification of Table 4.1.

3.6 Summary

This chapter laid the algebraic groundwork for treating classical combinatorial structures and their quantum analogues on equal footing. The guiding principle throughout has been that a finite set, once recast as a commutative C^*-algebra with a suitable functional, admits a natural non-commutative generalization, the quantum set, and that many further classical notion built on top of a set generalizes correspondingly. We saw this principle realized concretely: matrix algebras arose as crossed products of classical quantum sets, classical voltage and derived graphs embedded as a special (trivial-action) case of their quantum counterparts, and Gromada's correspondence translated quantum graphs on $M_n(\mathbb{C})$ into linear-algebraic data, i.e. subspaces of $\mathfrak{gl}(n, \mathbb{C})$, which are amenable to direct computation. Quantum isomorphisms, which are a generalisation of regular isomorphisms, will help us relate classical graphs to purely quantum ones in the next chapter.

Figure 3.7 summarizes this dictionary between the classical and quantum sides, for reference throughout the remainder of the thesis.

classical concept	quantum generalization
finite set	quantum set (B, ψ)
bijection	quantum set isomorphism
directed graph, adjacency matrix A	quantum graph, subspace $V \subseteq \mathfrak{gl}(n, \mathbb{C})$
loop-free graph	$V \subseteq \mathfrak{sl}(n)$
undirected graph	self-adjoint subspace
graph isomorphism	$V_1 = UV_2U^*$ for some $U \in U(n)$
group action	(dual) group action $\widehat{\alpha}\colon \widehat{G} \to \mathrm{Aut}(B, \psi)$
voltage graph (Γ, λ)	voltage quantum graph $\widetilde{A} = (\widetilde{A}_g)_{g \in G}$
derived graph Γ^λ	derived quantum graph $\widetilde{A} \rtimes_{\widehat{\alpha}} \widehat{G}$
intersection of graphs	$m(A_1 \otimes A_2)m^*$
isomorphism	quantum isomorphism

Figure 3.7: A dictionary between classical combinatorial notions and their quantum counterparts, as developed throughout this thesis.

Chapter 4

Quantum graphs over 3×3 matrices

The classification of quantum graphs on 2×2 matrices was carried out independently by Matsuda [11] and Gromada [6]. In this chapter, we attempt a partial classification of quantum graphs on 3×3 matrices. This case turns out to be considerably harder than the 2×2 case, and a classification of comparable simplicity is likely out of reach altogether, as we show in Lemma 4.5.1. We conclude the chapter by comparing our results with those of Hayes et al. [10].

4.1 Pauli matrices and their generalisations

4.1.1 Pauli matrices

An important basis of the Lie algebra $\mathfrak{sl}(2)$ is given by the Pauli matrices

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The Pauli matrices are Hermitian, traceless, and linearly independent;
since $\dim_{\mathbb{C}} \mathfrak{sl}(2) = 3$ by 2.3.3, they form a basis of $\mathfrak{sl}(2)$. They were already encountered in Example 2.3.2.

Because the Pauli matrices are Hermitian rather than skew-Hermitian, they do not themselves belong to $\mathfrak{su}(2)$. Instead, it is the matrices

$$i\sigma_1, \qquad i\sigma_2, \qquad i\sigma_3$$

that form a basis of $\mathfrak{su}(2)$, being both traceless and skew-Hermitian.

4.1.2 Generalised Pauli matrices

In dimensions greater than two, the role of the Pauli matrices is played by the *generalised Pauli matrices*, also known as the *clock* and *shift* matrices. In dimension three, they are defined by

$$P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \omega & 0 \\ 0 & 0 & \omega^2 \end{pmatrix}, \qquad Q = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix},$$

where $\omega = \exp(2\pi i/3)$.

The matrices P and Q are unitary and satisfy

$$P^3 = Q^3 = I,$$

together with the commutation relation

$$PQ = \omega\, QP.$$

The clock and shift matrices are related by the discrete Fourier transform. The Fourier matrix is

$$F = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{pmatrix},$$

which is unitary, i.e. satisfies $F^*F = FF^* = I$, and exchanges the roles of the clock and shift operators according to

$$FPF^* = Q^{-1}, \qquad FQF^* = P.$$

Consequently, P and Q are unitarily equivalent and represent two complementary bases of the three-dimensional Hilbert space. This duality is the finite-dimensional analogue of the relation between position and momentum operators, and is a fundamental feature of the generalised Pauli matrices.

To simplify later calculations, we first establish the following auxiliary lemma.

Lemma 4.1.1. *For all $a, b \in \mathbb{Z}$, the generalised Pauli matrices P, Q satisfy*

$$Q^a P^b = \omega^{-ab}\, P^b Q^a.$$

Proof. We first establish the base case $a = b = 1$, namely $QP = \omega^{-1}PQ$, and then extend to general a, b by induction.

For the base case, apply both sides to a basis vector e_i. Since $Pe_i = \omega^i e_i$,

$$(QP)e_i = Q(\omega^i e_i) = \omega^i\, e_{i+1},$$

while, since $Qe_i = e_{i+1}$ and $Pe_{i+1} = \omega^{i+1} e_{i+1}$,

$$(PQ)e_i = Pe_{i+1} = \omega^{i+1}\, e_{i+1}.$$

Comparing the two, $(QP)e_i = \omega^{-1}(PQ)e_i$ for every i, so $QP = \omega^{-1}PQ$, which is exactly $Q^1P^1 = \omega^{-1\cdot 1}P^1Q^1$.

We now show $Q^aP = \omega^{-a}PQ^a$ for every $a \in \mathbb{Z}$, by induction on a. The case $a = 0$ is trivial, and $a = 1$ is the base case above. For the forward inductive step, suppose $Q^aP = \omega^{-a}PQ^a$ holds; then

$$Q^{a+1}P = Q(Q^aP) = Q(\omega^{-a}PQ^a) = \omega^{-a}(QP)Q^a = \omega^{-a}(\omega^{-1}PQ)Q^a = \omega^{-(a+1)}PQ^{a+1},$$

using the base case $QP = \omega^{-1}PQ$ in the second-to-last step. For the backward inductive step, we multiply the base equation $QP = \omega^{-1}PQ$ on the left and right by Q^{-1} to obtain $PQ^{-1} = \omega^{-1}Q^{-1}P$, which rearranges to $Q^{-1}P = \omega PQ^{-1}$. Suppose $Q^aP = \omega^{-a}PQ^a$ holds; then

$$Q^{a-1}P = Q^{-1}(Q^aP) = Q^{-1}(\omega^{-a}PQ^a) = \omega^{-a}(Q^{-1}P)Q^a = \omega^{-a}(\omega PQ^{-1})Q^a = \omega^{-(a-1)}PQ^{a-1}.$$

This proves $Q^aP = \omega^{-a}PQ^a$ for all $a \in \mathbb{Z}$.

Finally, we extend to general b by an analogous induction on b, using the relation $Q^aP = \omega^{-a}PQ^a$ just established as the new base case. The case $b = 0$ is trivial. For the forward inductive step, suppose $Q^aP^b = \omega^{-ab}P^bQ^a$ holds; then

$$Q^aP^{b+1} = (Q^aP^b)P = \omega^{-ab}P^b(Q^aP) = \omega^{-ab}P^b(\omega^{-a}PQ^a) = \omega^{-a(b+1)}P^{b+1}Q^a,$$

completing the forward induction. For the backward inductive step, we multiply the relation $Q^aP = \omega^{-a}PQ^a$ on the left and right by P^{-1} to obtain $P^{-1}Q^a = \omega^{-a}Q^aP^{-1}$, meaning $Q^aP^{-1} = \omega^aP^{-1}Q^a$. Suppose $Q^aP^b = \omega^{-ab}P^bQ^a$ holds; then

$$Q^aP^{b-1} = (Q^aP^b)P^{-1} = \omega^{-ab}P^b(Q^aP^{-1}) = \omega^{-ab}P^b(\omega^aP^{-1}Q^a) = \omega^{-a(b-1)}P^{b-1}Q^a,$$

completing the induction. Hence $Q^aP^b = \omega^{-ab}P^bQ^a$ for all $a, b \in \mathbb{Z}$. □

Lemma 4.1.2. *The matrices P^aQ^b, $a, b \in \{0, 1, 2\}$, form an orthogonal basis of $M_3(\mathbb{C})$ with respect to the Hilbert–Schmidt inner product $\langle A, B\rangle = \mathrm{Tr}(A^*B)$.*

Proof. Since $\dim_{\mathbb{C}} M_3(\mathbb{C}) = 9$ and there are exactly $3^2 = 9$ matrices of the form P^aQ^b for $a, b \in \{0, 1, 2\}$, it suffices to show that they are pairwise orthogonal.

Recall that P and Q are unitary matrices, so $P^* = P^{-1}$ and $Q^* = Q^{-1}$. This gives $(P^aQ^b)^* = (Q^b)^*(P^a)^* = Q^{-b}P^{-a}$. The Hilbert–Schmidt inner product expands as

$$\langle P^aQ^b, P^cQ^d\rangle = \mathrm{Tr}\left((P^aQ^b)^*P^cQ^d\right) = \mathrm{Tr}\left(Q^{-b}P^{-a}P^cQ^d\right) = \mathrm{Tr}\left(Q^{-b}P^{c-a}Q^d\right).$$

Because Lemma 4.1.1 holds for all integer exponents, we can directly apply it to the terms Q^{-b} and P^{c-a}:

$$Q^{-b}P^{c-a} = \omega^{-(-b)(c-a)}\,P^{c-a}Q^{-b} = \omega^{b(c-a)}\,P^{c-a}Q^{-b}.$$

Substituting this back into the trace yields

$$\langle P^aQ^b, P^cQ^d\rangle = \mathrm{Tr}\left(\omega^{b(c-a)}P^{c-a}Q^{-b}Q^d\right) = \omega^{b(c-a)}\,\mathrm{Tr}\left(P^{c-a}Q^{d-b}\right).$$

To evaluate this trace, we consider two cases. First, if $b \neq d$, the exponent $d-b \in \{-2,-1,1,2\}$, meaning Q^{d-b} is a nontrivial cyclic permutation matrix. Multiplying the diagonal matrix P^{c-a} by Q^{d-b} results in a matrix with zeros everywhere on the main diagonal, forcing $\mathrm{Tr}(P^{c-a}Q^{d-b}) = 0$.

Second, if $b = d$, then $Q^{d-b} = Q^0 = I$, and the inner product simplifies to

$$\langle P^aQ^b, P^cQ^b\rangle = \mathrm{Tr}(P^{c-a}).$$

Since $P^{c-a} = \mathrm{diag}(1, \omega^{c-a}, \omega^{2(c-a)})$, its trace is given by a geometric series:

$$\mathrm{Tr}(P^{c-a}) = 1 + \omega^{c-a} + \omega^{2(c-a)} = \begin{cases} 3, & a = c, \\ 0, & a \neq c. \end{cases}$$

(The latter case follows from the identity $1 + \omega + \omega^2 = 0$, which applies since $a \neq c$ implies $c - a \not\equiv 0 \pmod 3$).

Combining both cases, we conclude that

$$\langle P^aQ^b, P^cQ^d\rangle = 3\,\delta_{ac}\delta_{bd}.$$

Therefore, the nine matrices P^aQ^b are pairwise orthogonal and form an orthogonal basis for $M_3(\mathbb{C})$. □

Excluding the identity matrix from this basis leaves eight traceless matrices, which together span $\mathfrak{sl}(3)$. A basis of $\mathfrak{su}(3)$ is then obtained by passing to suitable skew-Hermitian linear combinations of these eight matrices, exactly as the passage from σ_i to $i\sigma_i$ produced a basis of $\mathfrak{su}(2)$ above. In this sense, the clock and shift matrices furnish a natural higher-dimensional generalisation of the Pauli matrices, suited to describing three-level quantum systems.

4.2 Classification of self-adjoint subspaces spanned by a subset of the generalised Pauli basis $\left\{P^aQ^b\right\}_{a,b\in\{0,1,2\}}$

By the results of the previous section, it suffices to focus on self-adjoint subspaces of $\mathfrak{sl}(3)$. We stress at the outset, however, that the classification carried out in this section concerns only *self-adjoint subspaces spanned by a subset of the generalised Pauli basis* $\{P^aQ^b\}_{a,b\in\{0,1,2\}}$. Equivalently, by Lemma 4.1.2, subspaces that are a direct sum of some of the adjoint-invariant blocks introduced below (and not an arbitrary self-adjoint subspace of $\mathfrak{sl}(3)$). Since $\mathfrak{sl}(3)$ is eight-dimensional, an arbitrary subspace need not be spanned by any subset of a fixed basis: for instance, $\mathrm{span}\{P+$

$Q\}$ is a perfectly legitimate one-dimensional self-adjoint subspace of $\mathfrak{sl}(3)$ that is not of this restricted form, and there are in fact continuous families of subspaces of each dimension that are not coordinate subspaces with respect to $\{P^aQ^b\}_{a,b\in\{0,1,2\}}$. The classification below should therefore be read as covering a specific, tractable subfamily of the loop-free, undirected quantum graphs on $M_3(\mathbb{C})$, rather than all of them.

4.2.1 Fundamental adjoint-invariant pairs

Since P and Q are unitary, we have $P^* = P^{-1}$ and $Q^* = Q^{-1}$, meaning the conjugate transpose of an arbitrary basis matrix is $(P^aQ^b)^* = Q^{-b}P^{-a}$. By Lemma 4.1.1, we can commute these matrices to obtain

$$Q^{-b}P^{-a} = \omega^{-(-b)(-a)}P^{-a}Q^{-b} = \omega^{-ab}P^{-a}Q^{-b}.$$

Because $P^3 = Q^3 = I$, we can rewrite the negative exponents modulo 3, yielding the relation

$$(P^aQ^b)^* = \omega^{-ab}\,P^{3-a}Q^{3-b}.$$

For any subspace spanned by a subset of $\{P^aQ^b\}$ to be self-adjoint, it must be closed under conjugate transposition. Consequently, whenever P^aQ^b is included in the generating set, its conjugate counterpart $P^{3-a}Q^{3-b}$ must also be included. This pairing naturally partitions the eight non-identity basis matrices into four disjoint, adjoint-invariant subspaces of dimension two:

$$\begin{aligned}
\mathcal{P}_1 &= \{P, P^2\} && \text{(purely diagonal block)}\\
\mathcal{P}_2 &= \{Q, Q^2\} && \text{(purely shift block)}\\
\mathcal{P}_3 &= \{PQ, P^2Q^2\} && \text{(first mixed block)}\\
\mathcal{P}_4 &= \{P^2Q, PQ^2\} && \text{(second mixed block)}
\end{aligned}$$

Any choice of $S \in P(\{1,2,3,4\})$ is valid and gives rise to a self-adjoint subspace, where $P(A)$ is the power set of A. For instance, the choice $S = \{1,3\}$ gives the four-dimensional subspace

$$W_{1,3} = \operatorname{span}(\mathcal{P}_1 \cup \mathcal{P}_3) = \operatorname{span}\{P, P^2, PQ, P^2Q^2\}.$$

4.2.2 Action of the discrete Fourier transform

Let F denote the 3×3 discrete Fourier transform matrix. Recall that, by 4.1.2,

$$FPF^* = Q^2, \qquad FQF^* = P.$$

Lemma 4.2.1. *On the fundamental blocks $\mathcal{P}_k$, conjugation by F induces a pair-swapping involution:*

$$F\mathcal{P}_1F^* = \mathcal{P}_2, \qquad F\mathcal{P}_2F^* = \mathcal{P}_1, \qquad F\mathcal{P}_3F^* = \mathcal{P}_4, \qquad F\mathcal{P}_4F^* = \mathcal{P}_3.$$

Proof. Since F is unitary and satisfies $FPF^* = Q^2$ and $FQF^* = P$, conjugation by F is a $*$-automorphism sending $P \mapsto Q^2$ and $Q \mapsto P$. Hence, for any integers a, b,

$$F(P^aQ^b)F^* = (FPF^*)^a(FQF^*)^b = Q^{2a}P^b.$$

Applying this to the basis matrices of $\mathcal{P}_1$, we have

$$FPF^* = Q^2, \qquad FP^2F^* = Q^4 = Q,$$

so $F\mathcal{P}_1F^* = \operatorname{span}\{Q^2, Q\} = \mathcal{P}_2$. Because conjugation by F^2 maps $P \mapsto Q^2 \mapsto P^2$ and $P^2 \mapsto Q \mapsto P$, it preserves the subspace $\mathcal{P}_1$. Therefore, applying $F(\cdot)F^*$ to both sides of $F\mathcal{P}_1F^* = \mathcal{P}_2$ yields $F\mathcal{P}_2F^* = \mathcal{P}_1$.

Similarly, applying the F-conjugation to the generators of $\mathcal{P}_3$ yields

$$F(PQ)F^* = Q^2P, \qquad F(P^2Q^2)F^* = Q^4P^2 = QP^2.$$

Using the commutation relation $Q^aP^b = \omega^{-ab}P^bQ^a$ from Lemma 4.1.1, we can rewrite $Q^2P = \omega^{-2}PQ^2 = \omega PQ^2$ and $QP^2 = \omega^{-2}P^2Q = \omega P^2Q$. Thus,

$$F(PQ)F^* = \omega\, PQ^2, \qquad F(P^2Q^2)F^* = \omega\, P^2Q.$$

Since scalar multiples do not affect the linear span,

$$F\mathcal{P}_3F^* = \operatorname{span}\{\omega PQ^2, \omega P^2Q\} = \operatorname{span}\{PQ^2, P^2Q\} = \mathcal{P}_4.$$

By the same logic as before, applying $F(\cdot)F^*$ once more gives $F\mathcal{P}_4F^* = \mathcal{P}_3$. $\square$

4.2.3 Subspace classification

Because the fundamental building blocks $\mathcal{P}_k$ are non-overlapping 2-element sets, every self-adjoint subspace spanned by a subset of $\{P^aQ^b\}_{a,b\in\{0,1,2\}}$ is formed by taking a direct sum of a subset of these blocks (including the empty set, which yields the zero subspace).

Proposition 4.2.1. *Every self-adjoint subspace $V \subseteq M_3(\mathbb{C})$ generated by P and Q is classified up to unitary conjugacy by its dimension $m = \dim(V)$, as recorded in Table 4.1.*

The proof will be postponed until later.

Corollary 4.2.1. Including the trivial subspace V_0, there exist precisely 16 self-adjoint subspaces generated by P and Q.

m	subspaces	count
$1, 3, 5, 7$	none exist	0
0	$V_0 = \{0\}$	1
2	$V_k = \mathrm{span}(\mathcal{P}_k),\ k \in \{1, 2, 3, 4\}$	4
4	$W_{i,j} = \mathrm{span}(\mathcal{P}_i \cup \mathcal{P}_j),\ 1 \le i < j \le 4$	6
6	$U_k = \mathrm{span}\left(\bigcup_{i \neq k} \mathcal{P}_i\right),\ k \in \{1, 2, 3, 4\}$	4
8	$\mathfrak{sl}(3) = \mathrm{span}\left(\bigcup_{k=1}^{4} \mathcal{P}_k\right)$	1

Figure 4.1: Classification of self-adjoint subspaces $V \subseteq M_3(\mathbb{C})$ generated by P and Q by dimension.

This classification reduces to elementary combinatorics once the blocks $\mathcal{P}_1, \ldots, \mathcal{P}_4$ are fixed. Since these four blocks partition the eight non-identity Pauli generators into disjoint reflection-invariant pairs, choosing a self-adjoint subspace V of this restricted form amounts to nothing more than choosing a subset $S \subseteq \{1, 2, 3, 4\}$ of blocks to include, with

$$V = \mathrm{span}\left(\bigcup_{k \in S} \mathcal{P}_k\right).$$

There are $2^4 = 16$ such subsets in total, matching the count in the corollary. Moreover, since each included block contributes exactly two dimensions, the resulting dimension $\dim(V)$ is always even, explaining why V can never have odd dimension. The counts in each row of the table are then simply the binomial coefficients $\binom{4}{r}$, for $r = |S|$ blocks chosen, which is also why the table is symmetric under $m \leftrightarrow 8 - m$.

4.2.4 Full unitary equivalence within each dimension

The action of the discrete Fourier transform F considered above only produces 4 orbits among the six four-dimensional subspaces $W_{i,j}$, namely $\{W_{1,2}\}, \{W_{3,4}\}$, $\{W_{1,3}, W_{2,4}\}, \{W_{1,4}, W_{2,3}\}$. We now show that this is an artifact of restricting attention to F alone: once a second natural unitary is brought in, *all six* subspaces of dimension 4 turn out to be unitarily equivalent. More generally, all subspaces of any fixed dimension $m \in \{0, 2, 4, 6, 8\}$, among those generated by P and Q, are unitarily equivalent.

Consider the diagonal phase matrix

$$S = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \omega \end{pmatrix}.$$

Since S is diagonal, it commutes with P, so $SPS^{-1} = P$ and hence S fixes the block $\mathcal{P}_1 = \{P, P^2\}$. Its effect on Q is more interesting.

Lemma 4.2.2. *Conjugation by S cyclically permutes the blocks $\mathcal{P}_2, \mathcal{P}_3, \mathcal{P}_4$, while fixing $\mathcal{P}_1$. Explicitly,*

$$SQS^{-1} = \omega^{-1}PQ, \qquad SQ^2S^{-1} = P^2Q^2,$$
$$S(PQ)S^{-1} = \omega^{-1}P^2Q, \qquad S(P^2Q^2)S^{-1} = PQ^2.$$

Proof. Write e_1, e_2, e_3 for the standard basis, so that $Qe_i = e_{i+1}$ (indices mod 3) and $Se_i = s_ie_i$ with $s_1 = s_2 = 1$, $s_3 = \omega$. Then

$$(SQS^{-1})e_i = s_i^{-1}(SQ)e_i = s_i^{-1}s_{i+1}\, e_{i+1},$$

and computing the ratios s_{i+1}/s_i for $i = 1, 2, 3$ gives $1, \omega, \omega^2$ respectively. Comparing with $(PQ)e_i = p_{i+1}e_{i+1}$, where $p_1 = 1, p_2 = \omega, p_3 = \omega^2$, which gives coefficients $\omega, \omega^2, 1$, we see that SQS^{-1} has exactly the coefficients of PQ shifted by a global factor of ω^{-1}; that is, $SQS^{-1} = \omega^{-1}PQ$.

Squaring the second relation, and using the commutation relation $QP = \omega^{-1}PQ$ to simplify $(PQ)^2 = P(QP)Q = \omega^{-1}P^2Q^2$, we obtain

$$SQ^2S^{-1} = (SQS^{-1})^2 = \omega^{-2}(PQ)^2 = \omega^{-2} \cdot \omega^{-1}P^2Q^2 = P^2Q^2,$$

since $\omega^{-3} = 1$.

For the remaining two identities, use that S commutes with P:

$$S(PQ)S^{-1} = P\,(SQS^{-1}) = P \cdot \omega^{-1}PQ = \omega^{-1}P^2Q,$$
$$S(P^2Q^2)S^{-1} = P^2\,(SQ^2S^{-1}) = P^2 \cdot P^2Q^2 = P^4Q^2 = PQ^2,$$

using $P^3 = I$ in the last step. Collecting these four identities: S sends $\{Q, Q^2\} = \mathcal{P}_2$ onto $\{PQ, P^2Q^2\} = \mathcal{P}_3$ (up to phase), and sends $\mathcal{P}_3$ onto $\{P^2Q, PQ^2\} = \mathcal{P}_4$ (up to phase). By the same computation applied once more, or simply by counting (three blocks permuted among themselves by a single conjugation must be permuted cyclically or fixed, and we have already ruled out the latter), S also sends $\mathcal{P}_4$ onto $\mathcal{P}_2$. This proves the claim. □

In terms of the block indices $\{1, 2, 3, 4\}$, conjugation by F induces the permutation $\sigma_F = (1\,2)(3\,4)$ (as noted above), while conjugation by S induces the 3-cycle $\sigma_S = (2\,3\,4)$.

Remark 4.2.1. A basic fact from finite group theory will be used below: the alternating group A_4 (of order 12) has *no* subgroup of order 6. Indeed, a subgroup of order 6 would have index 2 in A_4 and hence be normal, but the only normal subgroups of A_4 are the trivial subgroup, the Klein four-group $V_4 = \{e, (12)(34), (13)(24), (14)(23)\}$, and A_4 itself, none of order 6.

Proposition 4.2.2. *The permutations σ_F and σ_S generate the full alternating group A_4 acting on the block indices $\{1, 2, 3, 4\}$. In particular this action is transitive on the six unordered pairs $\{i, j\} \subseteq \{1, 2, 3, 4\}$, i.e. for any unordered pairs $\{i, j\}$, $\{i', j'\}$ there is a permutation $\tau \in A_4$ such that $\tau(\{i, j\}) = \{i', j'\}$.*

Proof. Both σ_F (a double transposition) and σ_S (a 3-cycle) are even permutations, so the group H they generate is a subgroup of A_4. Because H contains an element of order 2 and an element of order 3, Lagrange's Theorem dictates that the order of H must be divisible by both 2 and 3. Since 2 and 3 are coprime, their least common multiple, 6, must also divide $|H|$. But A_4 has no subgroup of order 6 by Remark 4.2.1, and the only divisor of $|A_4| = 12$ that is a multiple of 6 and admits a subgroup is 12 itself; hence $H = A_4$.

It remains to verify that A_4 acts transitively on the set of six unordered pairs $P = \{\{i,j\} \subseteq \{1,2,3,4\} : i \neq j\}$. We first establish that A_4 is 2-transitive on $\{1,2,3,4\}$, meaning it acts transitively on the set $O = \{(i,j) \in \{1,2,3,4\}^2 : i \neq j\}$ of all $4 \times 3 = 12$ ordered distinct pairs. Consider the action of A_4 on O given by $\tau \cdot (i,j) = (\tau(i), \tau(j))$. The point-stabilizer of the ordered pair $(1,2)$ is:

$$\mathrm{Stab}_{A_4}((1,2)) = \{\tau \in A_4 : \tau(1) = 1 \text{ and } \tau(2) = 2\}.$$

Every non-identity element in A_4 is either a 3-cycle (which fixes exactly one element) or a product of two disjoint 2-cycles (which fixes zero elements). Consequently, no non-identity element of A_4 can fix two distinct points simultaneously, forcing $\mathrm{Stab}_{A_4}((1,2)) = \{e_{A_4}\}$.

By the Orbit-Stabilizer Theorem, the size of the orbit containing $(1,2)$ is:

$$|\operatorname{Orb}_{A_4}((1,2))| = \frac{|A_4|}{|\operatorname{Stab}_{A_4}((1,2))|} = \frac{12}{1} = 12.$$

Since the orbit size matches the total number of ordered pairs $|O| = 12$, A_4 acts transitively on ordered pairs.

Finally, for any two unordered pairs $\{i,j\}, \{i',j'\} \in P$, choosing ordering representatives (i,j) and (i',j') in O yields a permutation $\tau \in A_4$ such that $\tau(i) = i'$ and $\tau(j) = j'$. Thus:

$$\tau(\{i,j\}) = \{\tau(i), \tau(j)\} = \{i', j'\},$$

which proves that A_4 is transitive on unordered pairs. □

Proof of Proposition 4.2.1. By Proposition 4.2.2, for any two pairs $\{i,j\}, \{i',j'\} \subseteq \{1,2,3,4\}$ there is a permutation $\tau \in A_4$ with $\tau(\{i,j\}) = \{i',j'\}$, realized as a word in σ_F, σ_S and their inverses. Replacing each σ_F or σ_S in this word by the corresponding unitary F or S yields a unitary U with $U\mathcal{P}_i U^* = \mathcal{P}_{\tau(i)}$ for every block index i (up to scalar phases on individual generators, which do not affect the spans), and hence

$$UW_{i,j}U^* = U\big(\operatorname{span}(\mathcal{P}_i \cup \mathcal{P}_j)\big)U^* = \operatorname{span}(\mathcal{P}_{\tau(i)} \cup \mathcal{P}_{\tau(j)}) = W_{i',j'}.$$

The same argument, using only the transitivity of A_4 on single points, gives the case $m = 2$; the case $m = 6$ follows by taking orthogonal complements, since U_k is precisely the orthogonal complement of V_k inside $\mathfrak{sl}(3)$, and unitary conjugation commutes with taking orthogonal complements. □

4.3 Loop-free and undirected graphs on $M_3(\mathbb{C})$ from the generalised Pauli basis

In this section we exhibit, for each possible number of quantum edges $m \in \{0, 2, 4, 6, 8\}$, a representative quantum graph among those classified in Section 4.2, i.e. among the loop-free, undirected quantum graphs on $M_3(\mathbb{C})$ whose Gromada subspace is a direct sum of the blocks $\mathcal{P}_1, \ldots, \mathcal{P}_4$, realized as a derived quantum graph of a voltage quantum graph. As already emphasized in Section 4.2, this covers only part of the loop-free, undirected quantum graphs on $M_3(\mathbb{C})$; it is *not* a classification of all such graphs. Thanks to a result by Schäfer and Tobolski [15], namely Theorem 3.4.1, those graphs are quantum isomorphic to quantum graphs on $(\mathbb{C}^n \rtimes_{\hat{\alpha}} \hat{\mathbb{Z}}_n, \psi_{\rtimes})$, which, in turn, is isomorphic to $(M_3(\mathbb{C}), 3\,\mathrm{Tr})$ by Lemma 3.1.2, and so can be thought of as graphs on $(M_3(\mathbb{C}), 3\,\mathrm{Tr})$. By 3.4.1, all those graphs are quantum isomorphic to classical graphs on 9 vertices.

Theorem 4.3.1. *Among the non-empty, loop-free, undirected quantum graphs on* $(M_3(\mathbb{C}), 3\,\mathrm{Tr})$ *whose Gromada subspace is spanned by a subset of the generalised Pauli basis* $\{P^a Q^b\}_{a,b \in \{0,1,2\}}$*, there is exactly one isomorphism class in each non-trivial dimension* $m \in \{2, 4, 6, 8\}$*. In each class, there is a unique quantum graph that can be expressed as a derived quantum graph of a voltage quantum graph that can be identified with a classical voltage graph via 3.3.2.*

The adjacency matrices for those four derived quantum graphs, expressed in the basis of generalised Pauli matrices using the formula from Theorem 3.5.1, are presented in 4.2.

The trivial case $m = 0$, corresponding to the empty graph, is omitted, since it carries no meaningful structure to discuss.

The matrices in Proposition 4.3.1 were obtained by direct computation. For each subspace generated by P and Q as described in Table 4.1, we calculated the appropriate matrix using the formula given in Theorem 3.5.1 and compared them with matrices describing derived quantum graphs coming from quantum voltage graphs $\widetilde{A} = \{A_k\}_{k \in \hat{\mathbb{Z}}_3}$ given by the formula

$$\widetilde{A} \rtimes_{\hat{\alpha}} \hat{\mathbb{Z}}_3 = \sum_{k \in \hat{\mathbb{Z}}_3} m\big(X_k \otimes E_{\rtimes}^* \widetilde{A}_k E_{\rtimes}\big)\, m^*.$$

The following figures depict the voltage graphs and derived graphs that are quantum isomorphic to the representative graphs given as matrices in Figure 4.2. Concretely, the voltage graph with only self-loops (labeled $1, 2$ at every vertex) corresponds to the case $m = 2$; adding a 1-voltage triangle and a 2- voltage triangle going in the opposite direction to these loops gives $m = 4$; making the triangles undirected gives $m = 6$; and the voltage graph with edges of every voltage $0, 1, 2$ between every pair of distinct vertices and loops labeled with $1, 2$ gives $m = 8$, the complete quantum graph. All those derived graphs are undirected.

$$\begin{pmatrix} 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 \end{pmatrix}$$

Matrix of the graph Γ_2 obtained from $V_1 (m = 2)$

$$\begin{pmatrix} 2 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & -1 & 0 & 0 & 0 & \omega^2 & \omega & 0 & 0 \\ 0 & 0 & -1 & \omega & 0 & 0 & 0 & \omega^2 & 0 \\ 0 & 0 & \omega^2 & -1 & 0 & 0 & 0 & \omega & 0 \\ 1 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 \\ 0 & \omega & 0 & 0 & 0 & -1 & \omega^2 & 0 & 0 \\ 0 & \omega^2 & 0 & 0 & 0 & \omega & -1 & 0 & 0 \\ 0 & 0 & \omega & \omega^2 & 0 & 0 & 0 & -1 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 2 \end{pmatrix}$$

Matrix of the graph Γ_4 obtained from $W_{14} (m = 4)$

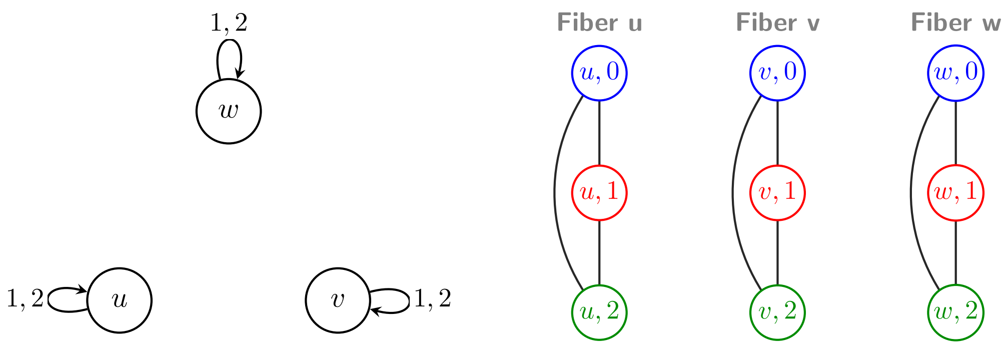
$$\begin{pmatrix} 2 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & -1 & 0 & 0 & 0 & -1 & -1 & 0 & 0 \\ 0 & 0 & -1 & -1 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & -1 & -1 & 0 & 0 & 0 & -1 & 0 \\ 1 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 \\ 0 & -1 & 0 & 0 & 0 & -1 & -1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & -1 & -1 & 0 & 0 \\ 0 & 0 & -1 & -1 & 0 & 0 & 0 & -1 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 2 \end{pmatrix}$$

Matrix of the graph Γ_6 obtained from $U_2 (m = 6)$

$$\begin{pmatrix} 2 & 0 & 0 & 0 & 3 & 0 & 0 & 0 & 3 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 3 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 3 & 0 & 0 & 0 & 3 & 0 & 0 & 0 & 2 \end{pmatrix}$$

Matrix of the graph Γ_8 obtained from $\mathfrak{sl}_3 (m = 8)$

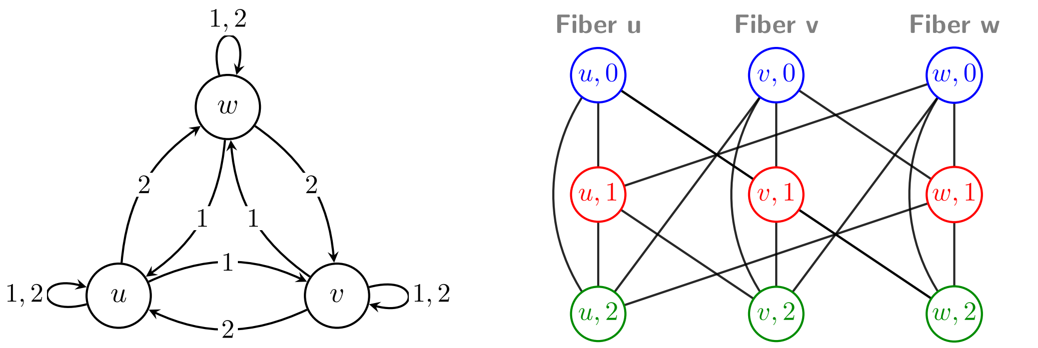
Figure 4.2: Adjacency matrices, expressed in the generalised Pauli basis via Theorem 3.5.1, of the four nontrivial loop-free and undirected quantum graphs on $(M_3(\mathbb{C}), 3\,\mathrm{Tr})$ up to isomorphism, one for each nontrivial dimension $m \in \{2, 4, 6, 8\}$.



Base Γ: directed loops

Derived Γ^λ: undirected 3-cycles

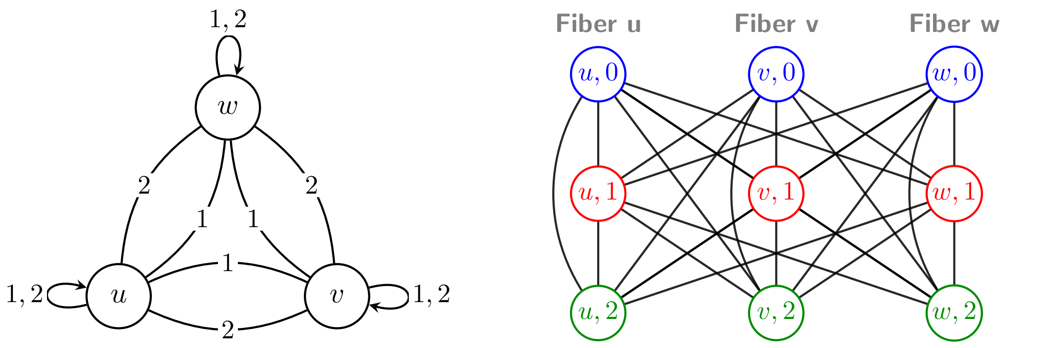
Figure 4.3: Voltage graph and derived cover for the loopfree, undirected quantum graph on $M_3(\mathbb{C})$ with $m = 2$ quantum edges.



Base Γ: directed triangle & loops

Derived Γ^λ: exactly 18 undirected edges

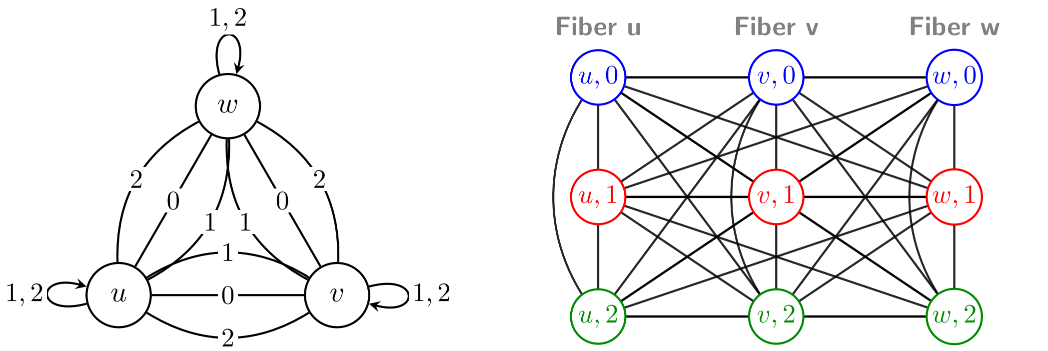
Figure 4.4: Voltage graph and derived cover for the loopfree, undirected quantum graph on $M_3(\mathbb{C})$ with $m = 4$ quantum edges.



Base Γ: directed multigraph

Derived Γ^λ: partial $K_{3,3}$ between fibers

Figure 4.5: Voltage graph and derived cover for the loopfree, undirected quantum graph on $M_3(\mathbb{C})$ with $m = 6$ quantum edges.



Base Γ: full directed multigraph

Derived Γ^λ: perfect $K_{3,3}$ between all fibers

Figure 4.6: Voltage graph and derived cover for the loopfree, undirected quantum graph on $M_3(\mathbb{C})$ with $m = 8$ quantum edges.

Since each of these voltage quantum graphs arises, via Theorem 3.4.1, as a derived quantum graph of an classical voltage graph, the discussion above in fact exhibits an explicit quantum isomorphism between each of the four representative quantum graphs of Theorem 4.3.1 and a classical derived graph on the nine vertices $\{u, v, w\} \times \mathbb{Z}_3$ shown on the right of each figure.

Proof of Theorem 4.3.1. By 4.2.1, every m-dimensional Gromada subspace is classified up to unitary conjugacy by its dimension. Consequently, 3.5.1 guarantees that the quantum graphs corresponding to these Gromada subspaces are isomorphic, which establishes the first statement.

Next, consider the graphs corresponding to the matrices in 4.2. Each of these can be obtained via the quantum voltage graph construction detailed in 3.3.7. Specifically, we take the quantum set $(\mathbb{C}^3, \psi_3)$ equipped with the action of the dual group $\widehat{\mathbb{Z}}_3 = \{\chi_0, \chi_1, \chi_2\}$ that cyclically rotates the coordinates. Explicitly, this action on a vector $z = (z_0, z_1, z_2) \in \mathbb{C}^3$ is given by

$$\chi_k \cdot (z_0, z_1, z_2) = (z_{0+k}, z_{1+k}, z_{2+k}),$$

where the addition is taken $\pmod 3$, so that the generator χ_1 yields $\chi_1 \cdot (z_0, z_1, z_2) = (z_1, z_2, z_0)$.

By combining this quantum set and group action with the appropriate classical voltage graphs, as illustrated in 4.3, 4.4, 4.5, and 4.6, we obtain the corresponding derived quantum graphs. By 3.1.2, each of these derived quantum graphs can be naturally treated as a graph on the quantum set $(M_3(\mathbb{C}), 3\,\mathrm{Tr})$. Finally, 3.4.1 ensures that the resulting quantum graphs are quantum isomorphic to the classical derived graphs depicted in the respective figures. □

Remark 4.3.1. We emphasize that this result applies exclusively to the representative graphs constructed in this section. Specifically, it concerns only those graphs whose Gromada subspace is spanned by sums of the blocks $\mathcal{P}_1, \ldots, \mathcal{P}_4$ introduced in Section 4.2. It does not extend to the full family of loop-free, undirected quantum graphs on $(M_3(\mathbb{C}), 3\,\mathrm{Tr})$, which is considerably larger. This distinction becomes immediately apparent when comparing our representatives with the complete classification of vertex-transitive quantum graphs on $M_3(\mathbb{C})$ by Hayes et al. [10], which we discuss in Section 4.4.

Remark 4.3.2. Although Theorem 4.3.1 establishes that our four representative graphs, as well as the empty graph representing $m = 0$, belong to five *distinct* classes under unitary conjugation, this does not guarantee they remain distinct under the strictly weaker relation of quantum isomorphism. Because the number of quantum edges, m, is not generally a quantum isomorphism invariant, representatives with different values of m could theoretically be quantum isomorphic. We do not investigate this question here, and we make no claim that the five classical graphs exhibited in this section represent distinct quantum isomorphism classes.

4.4 Vertex-transitive graphs

Classically, a graph is called vertex-transitive if, for any two vertices v, w, there exists a graph automorphism f such that $f(v) = w$. This notion was later extended to the quantum setting, and in their recent paper Hayes et al. [10] classify all vertex-transitive quantum graphs on $M_3(\mathbb{C})$ up to isomorphism.

Definition 4.4.1 (Vertex-transitive graphs, Hayes et al. [10], Definition 3.3)**.** A quantum graph A is called *vertex-transitive* if the set

$$\{u \in U(n) : uVu^* = V\}$$

spans $M_n(\mathbb{C})$, where $V \subseteq M_n(\mathbb{C})$ is the Gromada subspace of A.

The representative graphs obtained in Proposition 4.3.1 are all vertex-transitive, which is proven in the next Proposition 4.4.1. The proposition below establishes vertex-transitivity for the whole family of sixteen subspaces generated by P and Q, not merely the five representatives.

Proposition 4.4.1. *Every self-adjoint subspace $V \subseteq M_3(\mathbb{C})$ spanned by subsets of $\{P^aQ^b\}_{a,b\in\{0,1,2\}}$ gives rise to a vertex-transitive quantum graph.*

Proof. We first show that conjugation by any element P^aQ^b, $a, b \in \{0, 1, 2\}$, preserves each one-dimensional span$\{P^cQ^d\}$ individually. Using the commutation relation $PQ = \omega\, QP$, one computes

$$Q^bPQ^{-b} = \omega^{-b}P, \qquad P^aQP^{-a} = \omega^aQ,$$

and more generally, for any generator P^cQ^d,

$$(P^aQ^b)\, P^cQ^d\, (P^aQ^b)^{-1} = \omega^{\,bc-ad}\, P^cQ^d.$$

Thus conjugation by P^aQ^b fixes the generator P^cQ^d up to a scalar phase ω^{bc-ad}, and in particular sends span$\{P^cQ^d\}$ to itself.

Since every block $\mathcal{P}_k$ is a set of two such generators, and every self-adjoint subspace V is a direct sum of some subset of the blocks, it follows that

$$(P^aQ^b)\, V\, (P^aQ^b)^{-1} = V$$

for *every* $a, b \in \{0, 1, 2\}$, i.e. the full group $\{P^aQ^b\}_{a,b=0,1,2}$ stabilizes V, regardless of which subset of blocks was chosen.

But we have already shown that the nine matrices P^aQ^b form an orthogonal basis of $M_3(\mathbb{C})$ (Lemma 4.1.2); in particular they span $M_3(\mathbb{C})$. Since

$$\{P^aQ^b\}_{a,b=0,1,2} \;\subseteq\; \{u \in U(3) : uVu^* = V\},$$

the stabilizer of V already spans $M_3(\mathbb{C})$ on its own, so V is vertex-transitive. $\square$

Remark 4.4.1. In particular, this covers all sixteen subspaces of Table 4.1, including the extremal cases $V_0 = \{0\}$ and $\mathfrak{su}_3$, whose stabilizers are trivially all of $U(3)$. More strikingly, the proposition shows that vertex-transitivity holds already at the *minimal* nontrivial dimension $m = 2$: even the smallest quantum graphs $V_k = \mathrm{span}(\mathcal{P}_k)$ are vertex-transitive, stabilized by the same 9-element group that stabilizes every other subspace in the classification.

The following table shows the correspondences between the four representative quantum graphs of Theorem 4.3.1, together with the trivial graph, and the vertex-transitive graphs classified by Hayes et al. [10, Figure 1].

m	Quantum graphs from Figure 4.2	Hayes et al.
$m = 0$	the empty quantum graph	VT_0^3
$m = 2$	Γ_2	VT_2^3
$m = 4$	Γ_4	VT_4^3
$m = 6$	Γ_6	VT_6^3
$m = 8$	Γ_8	VT_8^3

Figure 4.7: Corresponding quantum graphs in 4.2 and Hayes et al.

Remark 4.4.2. Since Hayes et al. [10] classify *all* vertex-transitive quantum graphs on $M_3(\mathbb{C})$ up to isomorphism, not only those arising from subspaces generated by P and Q, their classification is a strict superset of the family constructed in Section 4.3. This gives independent confirmation that our five representative graphs do not exhaust the loop-free, undirected quantum graphs on $M_3(\mathbb{C})$, which is consistent with the restriction to the Pauli-coordinate family already noted in Section 4.2.

4.5 General classification of quantum graphs on $M_3(\mathbb{C})$

In this section, we examine the prospects for classifying quantum graphs without assuming them to be loop-free or undirected.

Most of the quantum graphs we have been dealing with have been quantum isomorphic to classical graphs. We will explicitly show a quantum graph on $M_3(\mathbb{C})$ that is not isomorphic to any classical graph.

Example 4.5.1 (Gromada [6], Example 8.5). The matrix

$$\lambda_8 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix},$$

which is one of the *Gell–Mann matrices*. Then the adjacency matrix $A = \lambda_8 \otimes \lambda_8$ defines a simple graph on $M_3(\mathbb{C})$. What's more, this quantum graph is not quantum isomorphic to any classical graph.

The proof that the quantum graph in Example 4.5.1 is not quantum isomorphic to any classical graph is non-elementary and is therefore omitted. We refer the interested reader to Gromada [6] for the details.

Theorem 4.5.1 demonstrates that simple quantum graphs on $M_2(\mathbb{C})$ can be fully classified, establishing that there are only finitely many such graphs up to isomorphism.

Theorem 4.5.1 (Gromada [6], Theorem 3.11)**.** *A simple quantum graph on $M_2(\mathbb{C})$ is determined up to isomorphism solely by the number of quantum edges $m \in \{0, 1, 2, 3\}$.*

The following proof is due to Gromada [6]. We repeat it here because we wish to point out a step that cannot be replicated in $M_3(\mathbb{C})$.

Proof. Recall first that $M_2(\mathbb{C})$ is assumed to carry no loops, which implies that a simple quantum graph on $(M_2(\mathbb{C}), \psi)$ admits at most $2^2 - 1 = 3$ quantum edges.

By Theorem 3.5.1, any simple quantum graph on $M_2(\mathbb{C})$ with m quantum edges is uniquely determined by an m-dimensional self-adjoint subspace $V \subseteq \mathfrak{sl}(2)$. By Lemma 3.5.2, it suffices to regard V as a real vector space, or equivalently, to identify iV with a subspace of the Lie algebra $\mathfrak{su}(2)$; by 2.3.3, we may then naturally view V as a real subspace $V \subset \mathbb{R}^3$.

By Theorem 3.5.1, two such subspaces $V, W \subset \mathbb{R}^3$ yield isomorphic quantum graphs if and only if they are related by conjugation by a unitary matrix $U \in SU(2)$. Under the identification of $\mathfrak{su}(2)$ with $\mathbb{R}^3$, this conjugation action of $SU(2)$ corresponds precisely to standard rotations, via the canonical isomorphism $PU(2) \cong SO(3)$.

Consequently, the quantum graphs corresponding to V and W are isomorphic if and only if V can be mapped onto W by a rotation in $SO(3)$. Since two real subspaces of $\mathbb{R}^3$ are congruent under $SO(3)$ if and only if they have the same dimension, the isomorphism class of the quantum graph is completely determined by $m = \dim V$. □

The proof of Theorem 4.5.1 relies on the well-known isomorphism $PU(2) \cong SO(3)$, which plays a key role throughout. This result does not generalise to higher dimensions, and in particular to dimension 3, as the following lemma shows.

Lemma 4.5.1. *The natural action*

$$PU(3) \curvearrowright \mathbb{P}(\mathbb{R}^8) = \{W \subseteq \mathbb{R}^8 : \dim(W) = 1\}$$

has infinitely many orbits.

Proof. Let $V = \mathfrak{su}(3) \cong \mathbb{R}^8$, and identify $\mathbb{P}(\mathbb{R}^8)$ with $\mathbb{P}(V)$. The action of $PU(3)$ on V is the adjoint representation

$$g \cdot X = gXg^{-1},$$

which induces an action on the 1-dimensional subspaces of V. It suffices to exhibit infinitely many pairwise non-conjugate lines inside a single explicit family, since this already forces infinitely many distinct orbits in $\mathbb{P}(V)$.

For $X \in V$, set

$$q(X) = -\operatorname{Tr}(X^2), \qquad c(X) = i \operatorname{Tr}(X^3).$$

Since the trace is invariant under conjugation,

$$q(gXg^{-1}) = q(X), \qquad c(gXg^{-1}) = c(X),$$

for every $g \in PU(3)$, and, for every $\lambda \in \mathbb{R}$,

$$q(\lambda X) = \lambda^2 q(X), \qquad c(\lambda X) = \lambda^3 c(X).$$

Consequently, whenever $q(X) \neq 0$, the ratio $c(X)^2/q(X)^3$ is unchanged both by the adjoint action and by rescaling X, and so defines an invariant of the line $\operatorname{span}\{X\} \in \mathbb{P}(V)$ under the induced $PU(3)$-action.

Now consider the family

$$X_a = i \operatorname{diag}(a, 1, -a-1) \in V, \qquad a \in [0,1].$$

A direct computation gives

$$q(X_a) = 2(a^2+a+1), \qquad c(X_a) = -3a(a+1),$$

and since $a^2 + a + 1 = (a + \frac{1}{2})^2 + \frac{3}{4} > 0$ for every $a \in [0,1]$, we have $q(X_a) \neq 0$ throughout this range. This lets us define, on the restricted domain $a \in [0,1]$, the function

$$I\colon [0,1] \to \mathbb{R}, \qquad I(a) = \frac{c(X_a)^2}{q(X_a)^3} = \frac{9a^2(a+1)^2}{8(a^2+a+1)^3},$$

which is exactly the invariant above evaluated on the line $\operatorname{span}\{X_a\}$.

The function I is continuous and non-constant on $[0,1]$; for instance,

$$I(0) = 0, \qquad I(1) = \frac{1}{6},$$

so by the intermediate value theorem I takes infinitely many distinct values as $a$ranges over $[0,1]$. Since I is constant on each $PU(3)$-orbit meeting the family $\{\operatorname{span}\{X_a\} : a \in [0,1]\}$, distinct values of $I(a)$ force the corresponding lines $\operatorname{span}\{X_a\}$ to lie in distinct orbits. Hence this family alone meets infinitely many orbits, so the action of $PU(3)$ on $\mathbb{P}(\mathbb{R}^8)$ has infinitely many orbits. □

As Lemma 4.5.1 illustrates, the landscape of quantum graphs on $M_3(\mathbb{C})$ is vastly more complex than that of $M_2(\mathbb{C})$. Even in the simplest non-trivial case of a single quantum edge ($m = 1$), there exists an uncountable family of pairwise non-isomorphic quantum graphs parameterized by continuous invariants. Consequently, the discrete parameter m is entirely insufficient for classification in higher dimensions. A complete, general classification of quantum graphs on $M_3(\mathbb{C})$ would therefore require navigating a highly intricate space of $PU(3)$-orbits, making the problem significantly more difficult and fundamentally different from the finite classification established in Theorem 4.5.1.

Chapter 5

Conclusions and Further Research

In this thesis we set out to study quantum graphs on 3×3 matrices, building on the classification of quantum graphs on $M_2(\mathbb{C})$ obtained independently by Matsuda [11] and Gromada [6]. After recalling the necessary background on C^*-algebras, quantum sets, and the theory of voltage and derived quantum graphs due to Schäfer and Tobolski [15], we focused on the loop-free, undirected quantum graphs on $M_3(\mathbb{C})$ generated by the generalised Pauli matrices P and Q. We showed that, although the discrete Fourier transform alone only partitions the sixteen self-reflective subspaces generated by P and Q into ten orbits, a second natural unitary reveals that all subspaces of a fixed dimension are in fact unitarily equivalent, reducing the classification to exactly five isomorphism classes – one for each even dimension $m \in \{0, 2, 4, 6, 8\}$. We then exhibited each nontrivial class explicitly as the derived quantum graph of a classical voltage graph over $\mathbb{Z}_3$, thereby showing that every such quantum graph is quantum isomorphic to a classical graph on nine vertices, and proved that all of them are vertex-transitive.

We also examined how far the picture available in dimension two extends to dimension three. The clean classification of quantum graphs on $M_2(\mathbb{C})$ by the number of quantum edges alone rests on the exceptional isomorphism $PU(2) \cong SO(3)$, and we showed that this argument provably breaks down for $M_3(\mathbb{C})$: the action of $PU(3)$ on projective subspaces of $\mathfrak{su}(3)$ has infinitely many orbits, so infinite continuous families of mutually non-isomorphic quantum graphs exist already for a fixed number of quantum edges. We further exhibited a simple quantum graph on $M_3(\mathbb{C})$ that is not quantum isomorphic to any classical graph, in contrast to the graphs constructed from P and Q, and compared our results to the independent classification of vertex-transitive quantum graphs on $M_3(\mathbb{C})$ recently obtained by Hayes et al. [10].

The most immediate direction for further work is to extend the classification obtained here from $M_3(\mathbb{C})$ to M_n for general n. This appears to be considerably more difficult than the $n = 3$ case treated in this thesis: already the transition from $M_2(\mathbb{C})$ to $M_3(\mathbb{C})$ required abandoning the single-invariant classification enabled by $PU(2) \cong SO(3)$ in favour of a much finer, case-by-case analysis, and the loss of any comparable exceptional isomorphism for $PU(n)$, $n \geq 4$, suggests that a full classification for general n may require substantially new ideas rather than a direct generalisation of the methods used here. A more modest but tractable goal would be to carry out the analogue of our classification for M_4 or M_5, where the generalised Pauli matrices and the voltage graph techniques of this thesis should still apply, and where a comparison with the vertex-transitivity results of Hayes et al. [10] may again prove fruitful.

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